

Energy

Solar panels are composed of specially treated silicon.



Biochemistry

The flash of the firefly results from a chemical reaction in the insect.



Technology

OLEDs (organic light-emitting diodes) are used in high-end cell phone, tablet, and television displays.



Medicine

Chemists are constantly striving to design new and improved drugs for treating disease.

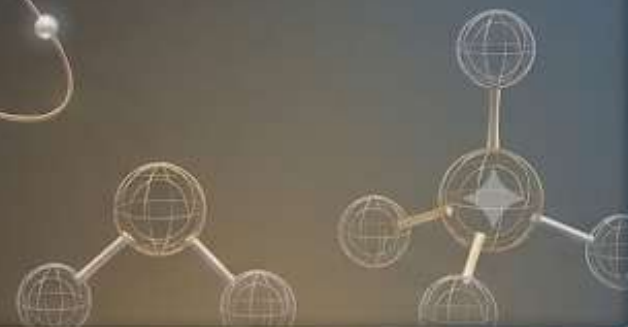
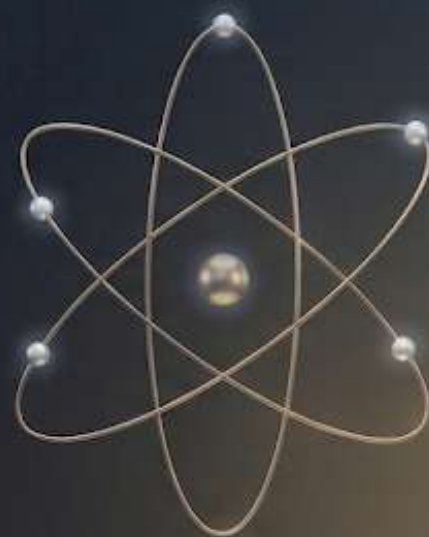


Chemistry

ASASIpintar

Topic 1A

Matter and Atomic Structure



Chapter Outline

- | | | | |
|----|---------------------------------------|----|-----------------------|
| 01 | Classification of matter | 06 | Chemical formula |
| 02 | Dalton's atomic theory | 07 | Chemical nomenclature |
| 03 | Structure of the atom | 08 | Molecular compounds |
| 04 | Atomic number, mass number & isotopes | 09 | Relative mass |
| 05 | Ions | 10 | Mass spectrometer |

Learning Objectives

- Classify matter as substances or mixtures; distinguish homogeneous from heterogeneous.
- Categorise substances as elements or compounds.
- Outline Dalton's atomic theory and the Law of Conservation of Mass.
- Describe the physical properties of electrons, protons and neutrons.
- Use atomic number and mass number to determine subatomic particle counts.
- Explain what isotopes are and compare their chemical vs physical properties.
- Identify cations and anions; determine formulas of ionic compounds.
- Name ionic, molecular, and oxyacid compounds using IUPAC nomenclature.
- Calculate relative atomic mass from mass spectrometry data.



SECTION ONE

01 Classification of Matter

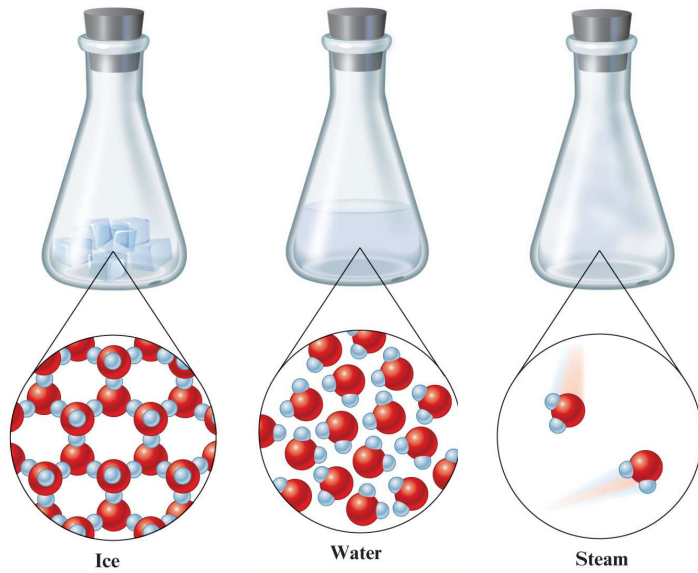
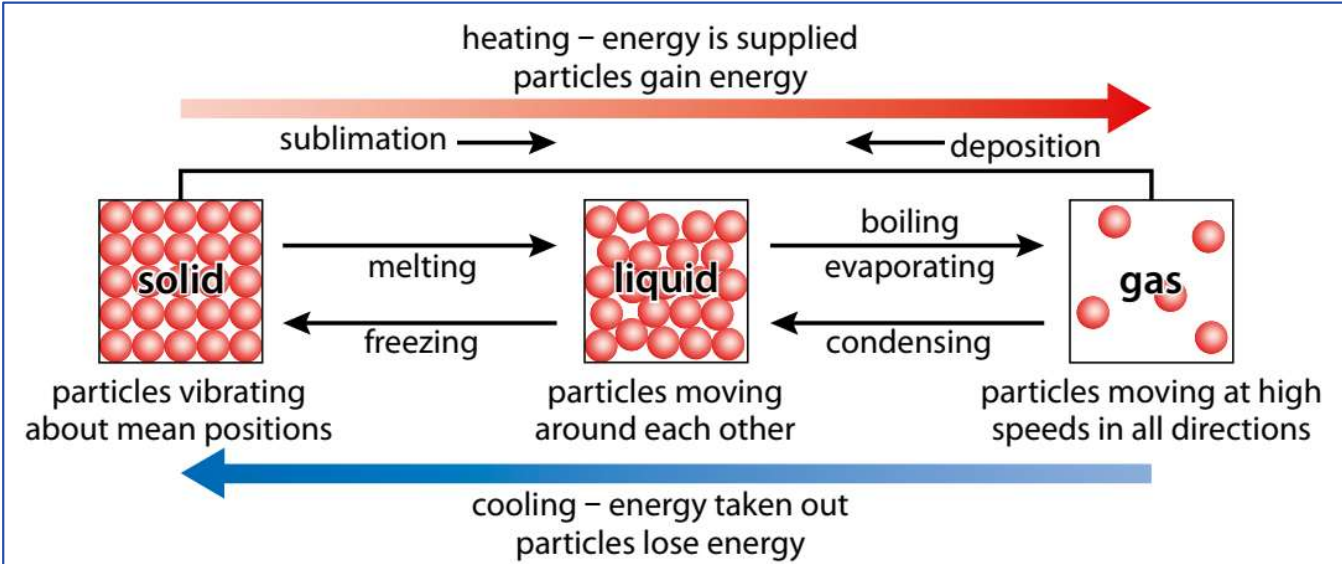
Anything that occupies space and has mass.

1.0 CLASSIFICATION OF MATTER

Matter



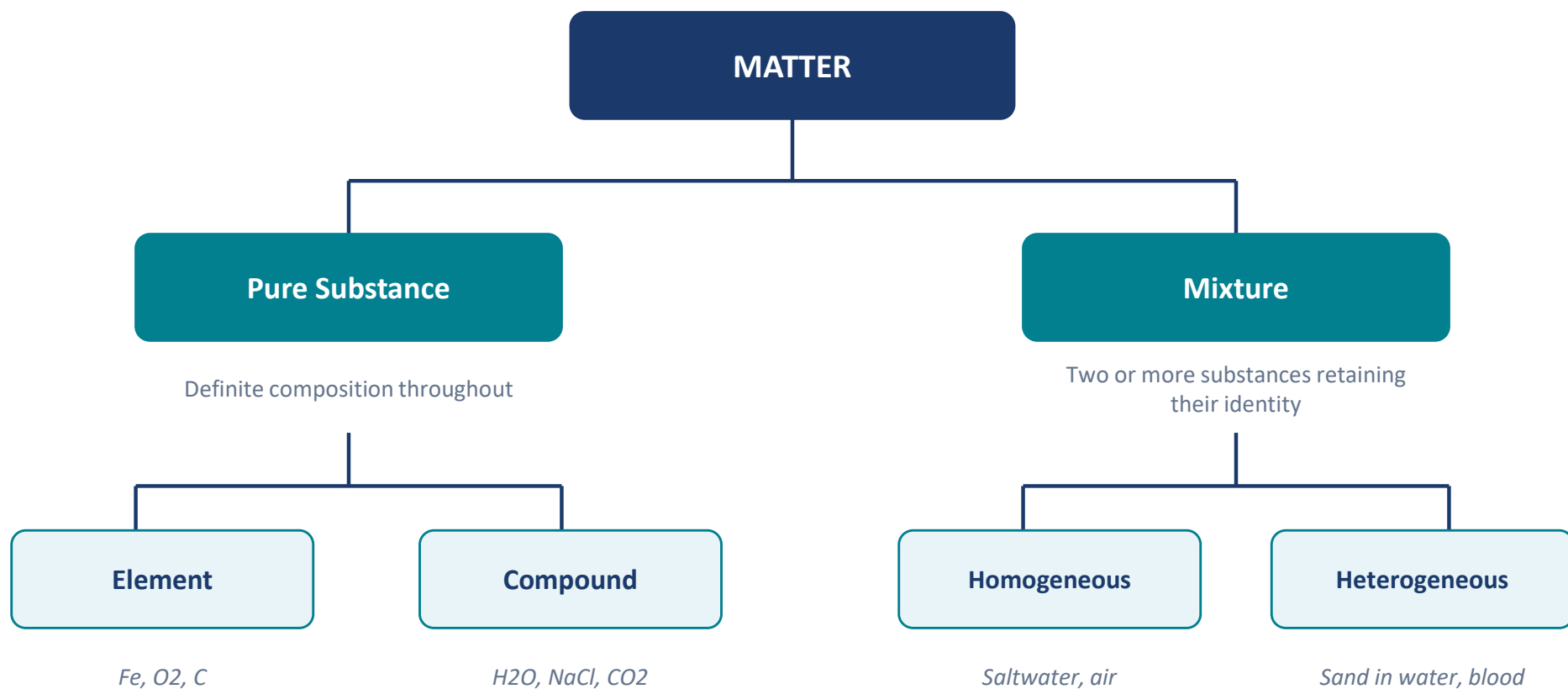
Anything that occupies space and has mass.



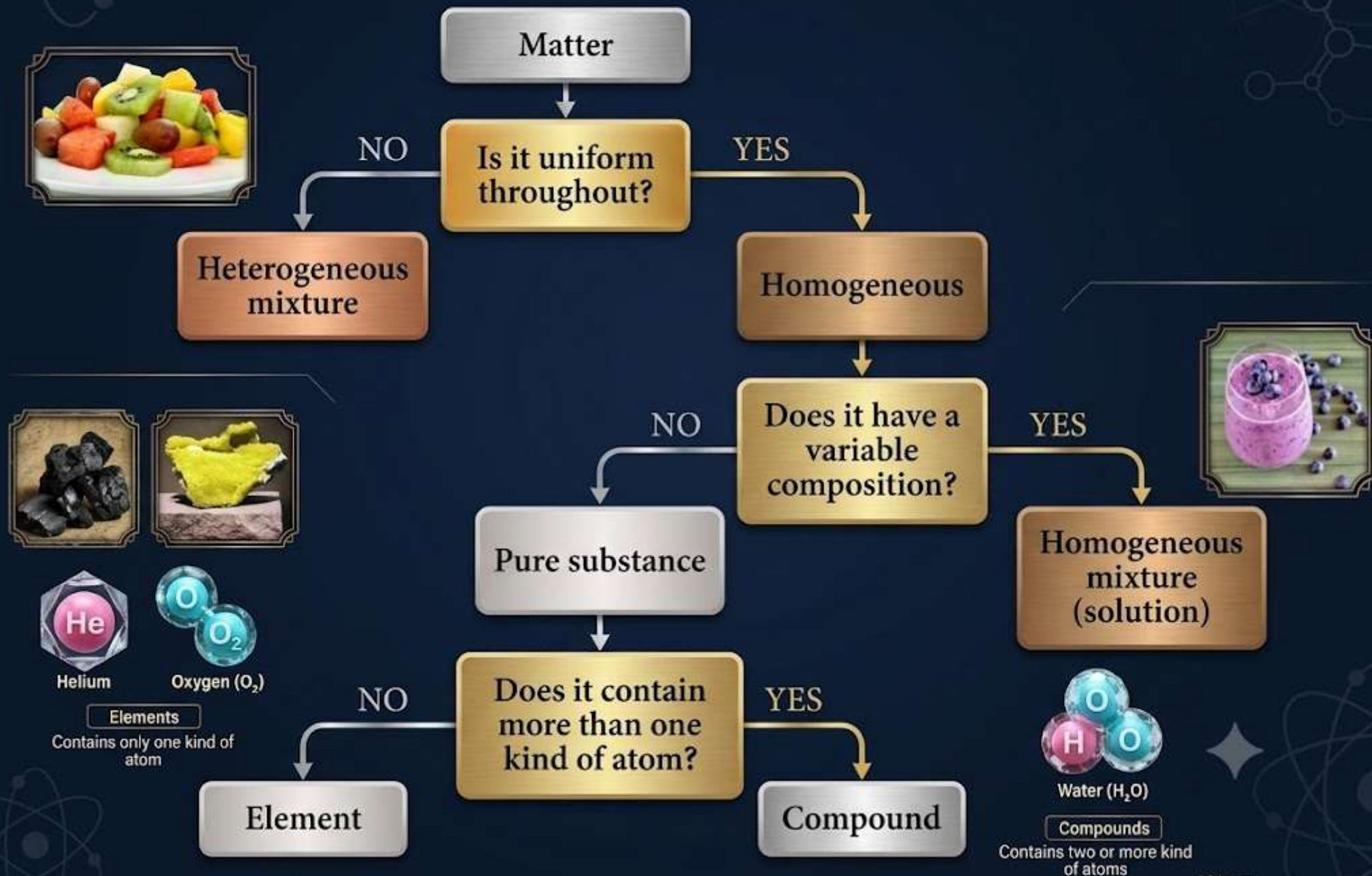
	Solids	Liquids	Gases
Distance between particles	close together	close but further apart than in solids	particles far apart
Arrangement	regular	random	random
Shape	fixed shape	no fixed shape – take up the shape of the container	no fixed shape – fill the container
Volume	fixed	fixed	not fixed
Movement	vibrate	move around each other	move around in all directions
Speed of movement	slowest	faster	fastest
Energy	lowest	higher	highest
Forces of attraction	strongest	weaker	weakest

The properties of the three states of matter.

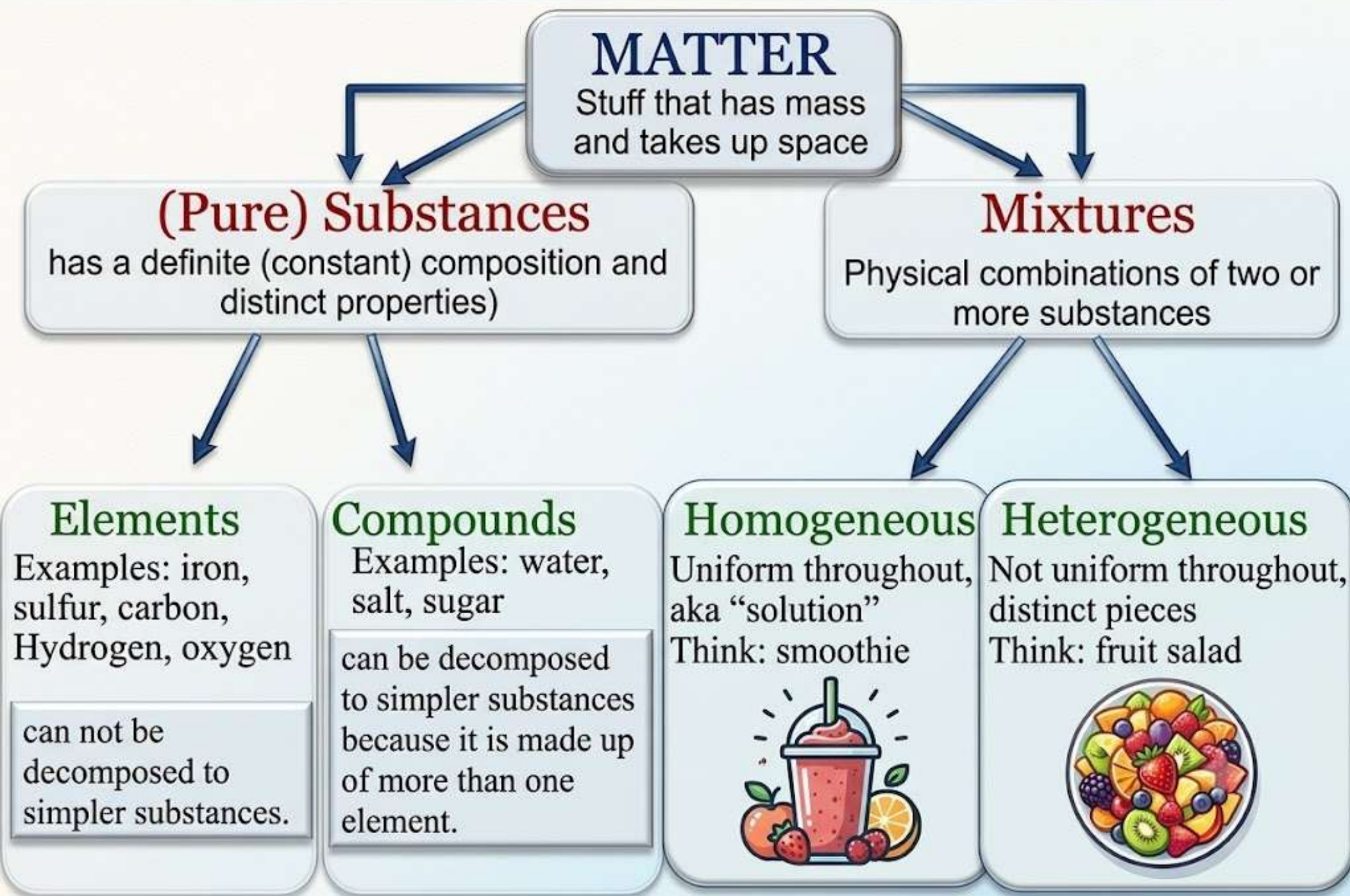
Classification of Matter



CLASSIFICATION OF MATTER BASED ON COMPOSITION



1.0 CLASSIFICATION OF MATTER



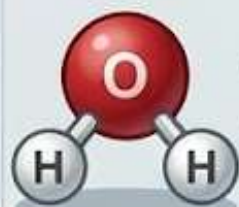
EXAMPLES OF PURE CHEMICAL COMPOUNDS (NOT MIXTURES)

(Hetero-atomic)
Compounds

Pure
Substances

(Homo-atomic)
Elements

Water (H_2O)



Two 'H' atoms bonded to one 'O' atom. Definite ratio, cannot be broken down by physical means.



Diatomic Oxygen (O_2)

Two oxygen atoms bonded together as a single substance. No other atom types.



Table Salt ($NaCl$)

A solid lattice of sodium (Na) and chlorine (Cl) ions. Fixed proportion.



Sulfur Crystal (S_8)

A molecular form of eight sulfur atoms. Pure elemental crystal.



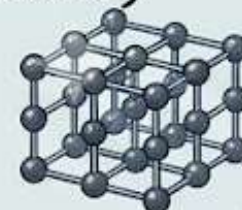
Methane (CH_4)

A natural gas molecule of and four hydrogen atoms. Specific bonding.



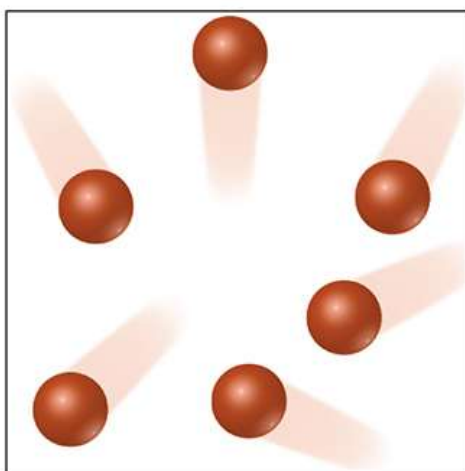
Diamond (pure Carbon)

A giant covalent lattice made entirely of carbon atoms. A single kind of atom.

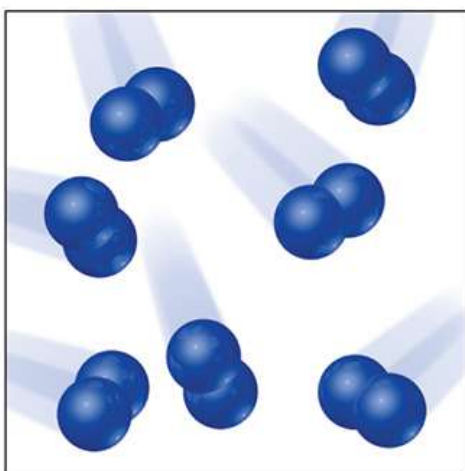


1.0 CLASSIFICATION OF MATTER

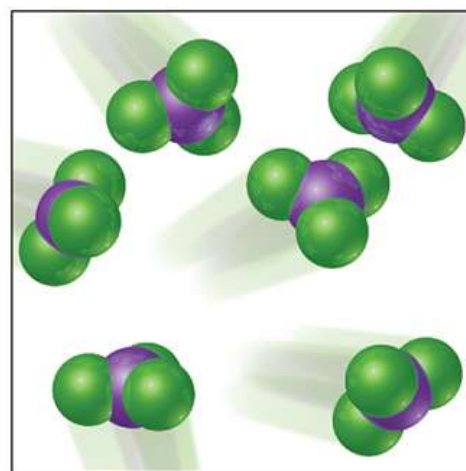
MATTER is anything that has mass and takes up space.



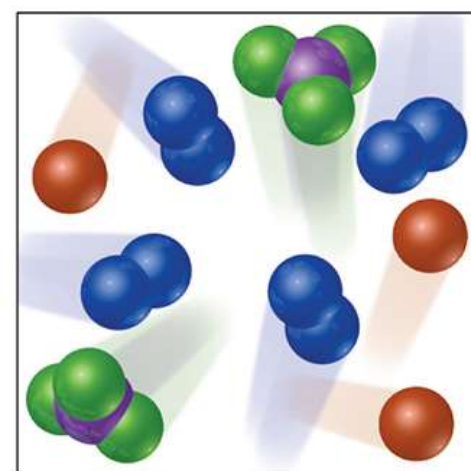
(a) Atoms of an element



(b) Molecules of an element



(c) Molecules of a compound

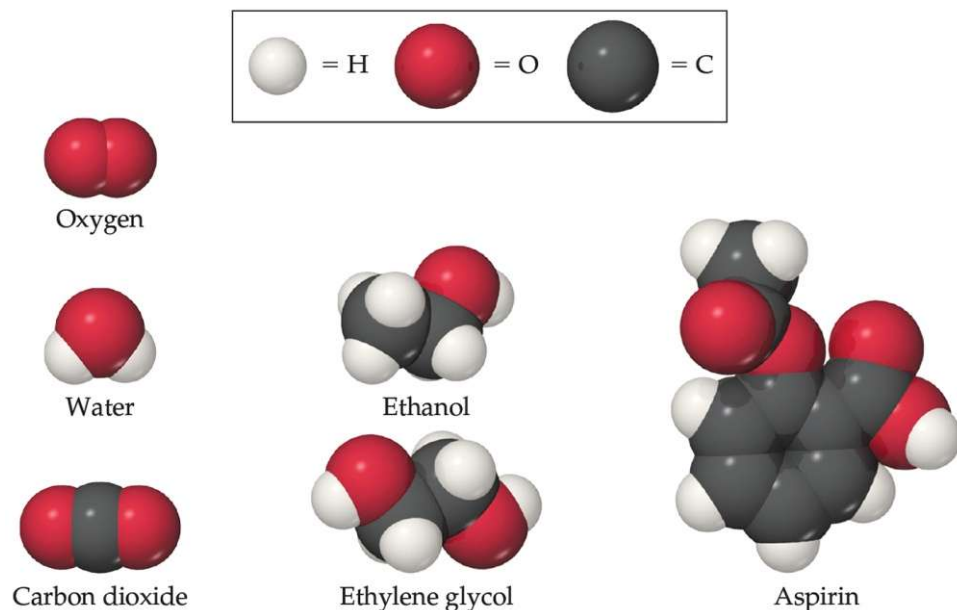


(d) Mixture of elements and a compound

Elements are composed of only one kind of atom.

Compounds must have at least two kinds of atoms.

CLASSIFICATION OF MATTER BASED ON COMPOSITION



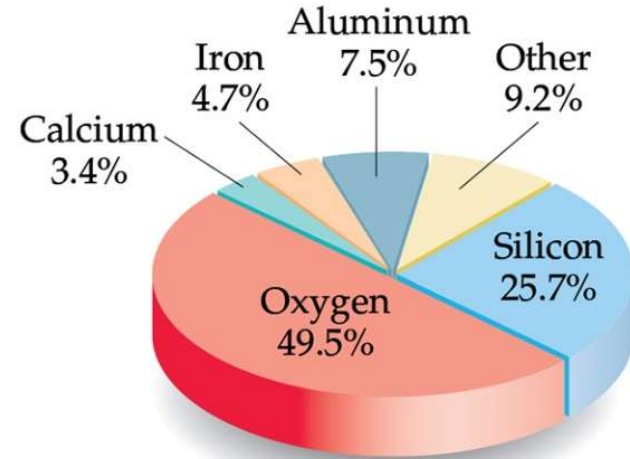
Note: Balls of different colors are used to represent atoms of different elements. Attached balls represent connections between atoms that are seen in nature. These groups of atoms are called **molecules**.

- **Atoms** are the building blocks of matter.
- Each **element** is made of a unique kind of atom but can be made of more than one atom of that kind.
- A **compound** is made of atoms from two or more different elements.

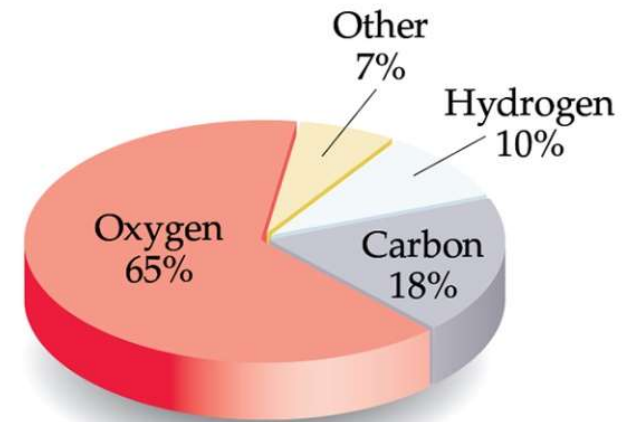
1.0 CLASSIFICATION OF MATTER

ELEMENTS AND COMPOSITION

- There are currently 118 named elements.
- Only five elements make up 90% of the earth's crust by mass.
- Only three elements make up 90% of the human body by mass!
- Note the importance of oxygen!



Earth's crust



Human body

2.0 DALTON'S ATOMIC THEORY (1808) Describe matter in terms of atoms

JOHN DALTON



DALTON'S ATOMIC THEORY (PROPOSED IN 1808):

- All elements are made of tiny atoms.
- Atoms cannot be subdivided.
- Atoms of the same element are exactly alike.
- Atoms of different elements can join to form molecules.



SECTION TWO

02 Dalton's Atomic Theory

All matter is built from indivisible atoms — proposed in 1808.

Dalton's Atomic Theory

1. Each element is composed of extremely small particles called atoms.



An atom of the element oxygen



An atom of the element nitrogen

2. All atoms of a given element are identical, but the atoms of one element are different from the atoms of all other elements.



Oxygen



Nitrogen

3. Atoms of one element cannot be changed into atoms of a different element by chemical reactions; atoms are neither created nor destroyed in chemical reactions.

Oxygen



Nitrogen

4. Compounds are formed when atoms of more than one element combine; a given compound always has the same relative number and kind of atoms.



+



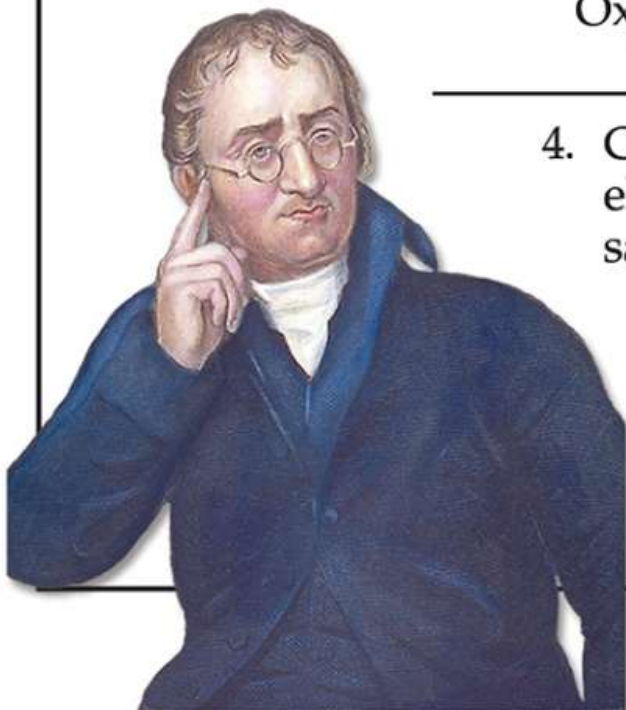
N

O

NO

Elements

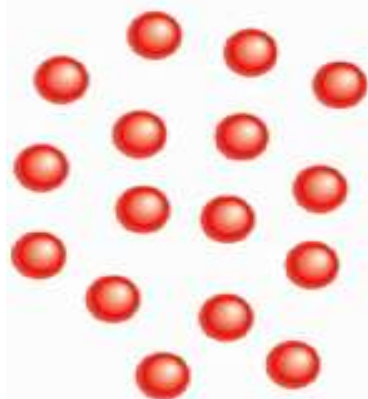
Compound



Law of Conservation of Mass - mass cannot be created or destroyed, although it may be rearranged.

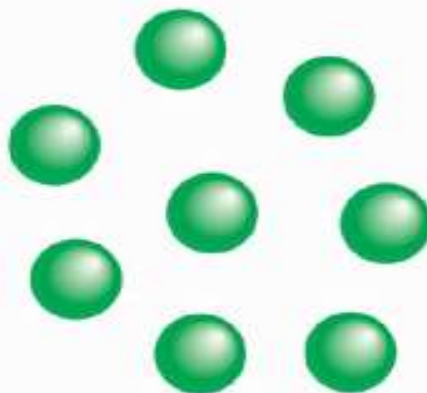
2.0 DALTON'S ATOMIC THEORY (1808)

Dalton's Atomic Theory

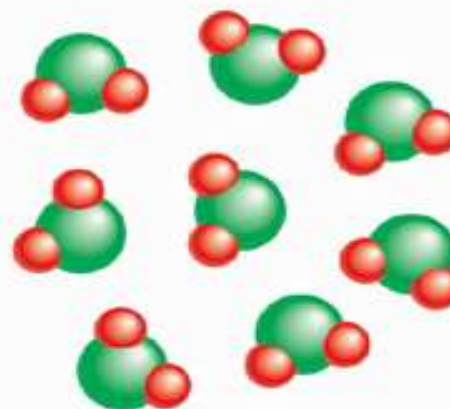


Atoms of element X

(a)



Atoms of element Y



Compound of elements X and Y

(b)

16 X

+

8 Y



8 X₂Y

Law of Conservation of Mass



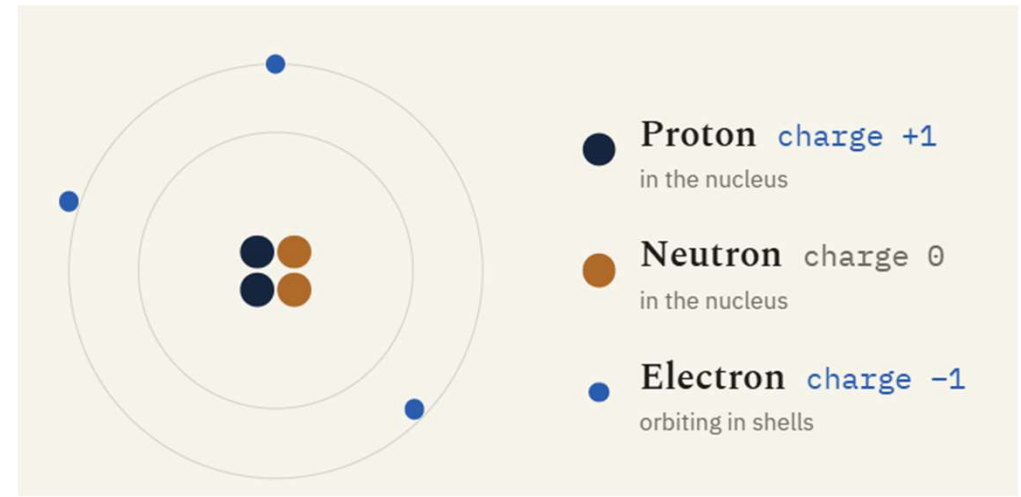
SECTION THREE

03 Structure of the Atom

A nucleus of protons and neutrons, surrounded by electrons.

3.0 THE STRUCTURE OF THE ATOM

- ☼ All neutral atoms can be identified by number of protons and neutrons.
- ☼ Proton has a positive charge.
- ☼ Neutron has no charge.
- ☼ Electron has a negative charge.



Particle	Mass	Charge*
Electron	9.109×10^{-31} kg	1 -
Proton	1.673×10^{-27} kg	1 +
Neutron	1.675×10^{-27} kg	None

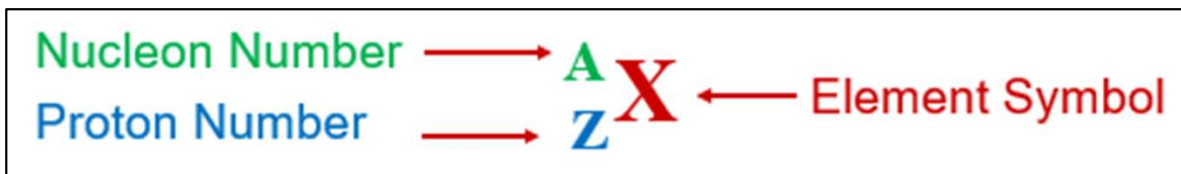
*The magnitude of the charge of the electron and the proton is 1.60×10^{-19} C.



SECTION FOUR

04 Atomic Number, Mass Number & Isotopes

4.0 ATOMIC NUMBER, MASS NUMBER AND ISOTOPES



● Nucleon number (A)

→ total number of proton and neutrons present in the nucleus of the atom of an element

→ also known as **mass number**

● Proton number (Z)

→ number of proton in the nucleus of the atom of an element

→ also known as **atomic number**



A • NUCLEON NUMBER (top)

Total protons + neutrons in the nucleus. Also called the *mass number*.

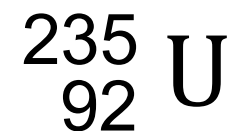
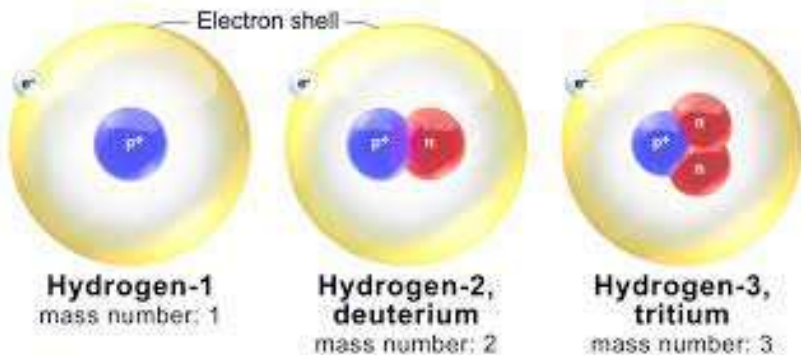
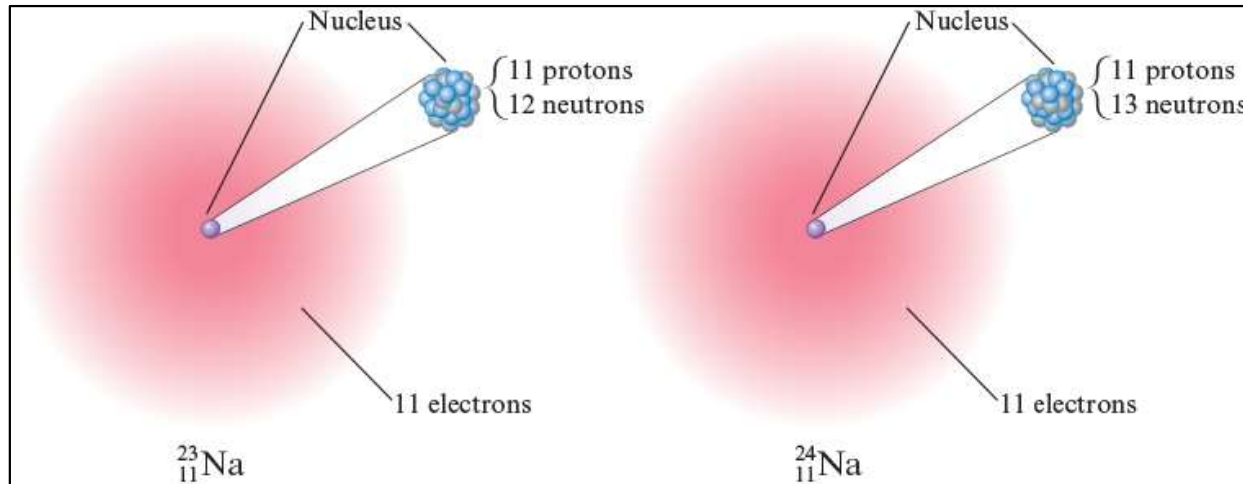
Z • PROTON NUMBER (bottom)

Number of protons in the nucleus. Also called the *atomic number*.

$$\text{neutrons} = A - Z = 23 - 11 = 12$$

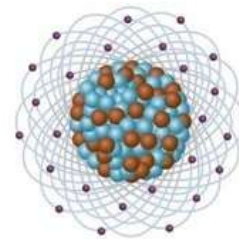
4.0 ATOMIC NUMBER, MASS NUMBER AND ISOTOPES

- Isotopes are **two or more atoms** of the **same element** that have the **same proton number** in their nucleus but **different nucleon number**.



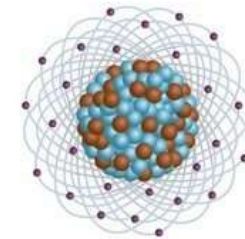
ISOTOPES

URANIUM-238



● 92 protons ● 146 neutrons
More neutrons—heavier

URANIUM-235



● 92 protons ● 143 neutrons
Fewer neutrons—lighter and less stable

4.0 ATOMIC NUMBER, MASS NUMBER AND ISOTOPES

Isotopes are different atoms of the same element with different mass numbers: i.e. different numbers of neutrons in the nucleus.

Isotope	Protons	Neutrons	Electrons
${}^1_1\text{H}$	1	0	1
${}^2_1\text{H}$	1	1	1
${}^3_1\text{H}$	1	2	1
${}^{12}_6\text{C}$	6	6	6
${}^{13}_6\text{C}$	6	7	6
${}^{14}_6\text{C}$	6	8	6
${}^{35}_{17}\text{Cl}$	17	18	17
${}^{37}_{17}\text{Cl}$	17	20	17

The numbers of subatomic particles in some common isotopes.

4.0 ATOMIC NUMBER, MASS NUMBER AND ISOTOPES

Isotopes have the same chemical properties (they react in exactly the same way) but different physical properties (e.g. different melting points and boiling points).

- Isotopes react in the same way because they have the same numbers of electrons,
- → chemical reactions depend only on the **number and arrangement of electrons** and
→ not on the composition of the nucleus.
- Example, both protium and deuterium would react in the same way with nitrogen:
$$\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3 \qquad \text{N}_2 + 3\text{D}_2 \rightleftharpoons 2\text{ND}_3$$
- Isotopes have different physical properties because,
 - different masses mean that their atoms move at different speeds.
 - The boiling point of $^1_1\text{H}_2$ is -253°C , whereas that of $^2_1\text{H}_2$ is -250°C .
 - Heavy water (D_2O) has a melting point of 3.8°C and a boiling point of 101.4°C

The isotopes of an element with fewer neutrons will have:

- | | | |
|-----------------|---|----------------------------------|
| Lower masses | • | faster rate of diffusion |
| Lower densities | • | lower melting and boiling points |



Properties of Isotopes

Chemically identical

Same number of electrons → reactions depend on electrons, not the nucleus.

Physically different

Different masses → atoms move at different speeds.

fewer neutrons →

faster diffusion · lower density · lower m.p./b.p.

4.0 ATOMIC NUMBER, MASS NUMBER AND ISOTOPES

Do You Understand Isotopes?

How many protons, neutrons, and electrons are in $^{14}_6\text{C}$?

6 protons, 8 (14 - 6) neutrons, 6 electrons

How many protons, neutrons, and electrons are in $^{11}_6\text{C}$?

6 protons, 5 (11 - 6) neutrons, 6 electrons

KEY IDEA

Neutrons = nucleon –
proton number. Isotopes
share protons & electrons;
only neutrons differ.

Which of the following statements are true?

I. The number of protons is the same for all neutral atoms of an element

II. The number of electrons is the same for all neutral atoms of an element

III. The number of neutrons is the same for all neutral atoms of an element

- A. I, II, and III are true
- B. Only I and II are true
- C. Only II and III are true
- D. Only I and III are true
- E. I, II, and III are false



SECTION FIVE

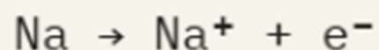
05 Ions

An atom that has lost or gained electrons, and so carries a charge.

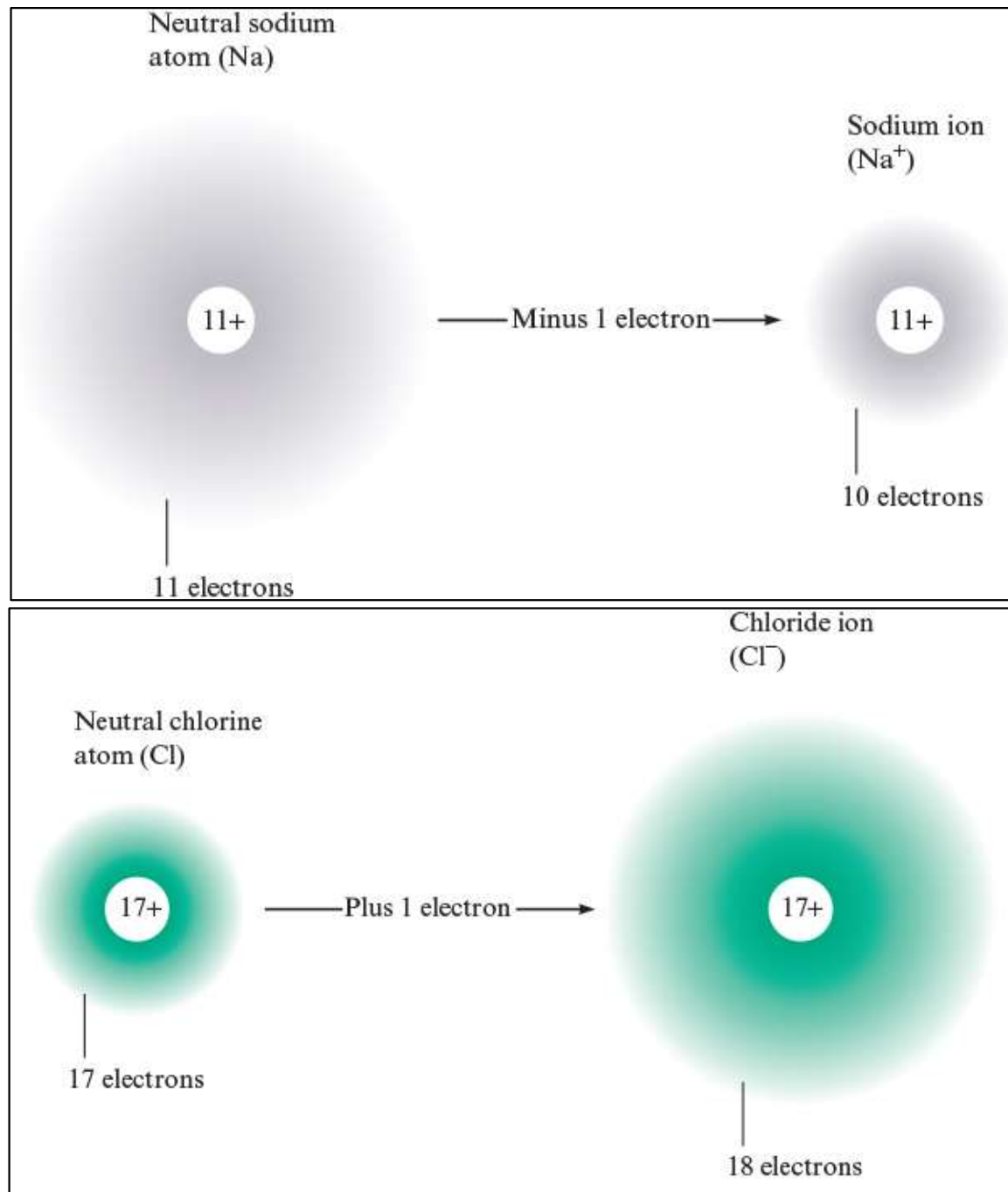
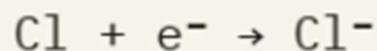
5.0 IONS

Ions are charged particles that are formed when an atom loses or gains (an) electron(s).

A **positive ion (cation)** is formed when an atom **loses** (an) **electron(s)** so that the ion has more protons(+) than electrons(-)



A **negative ion (anion)** is formed when an atom **gains** (an) **electron(s)** so that the ion has more electrons(-) than protons(+)



5.0 IONS (MONOATOMIC ION)

1																17	18
H ⁺	2															H ⁻	N O B L E G A S E S
Li ⁺																F ⁻	
Na ⁺	Mg ²⁺															Cl ⁻	
K ⁺	Ca ²⁺															Br ⁻	
Rb ⁺	Sr ²⁺															I ⁻	
Cs ⁺	Ba ²⁺																

Transition metals

13 14 15 16

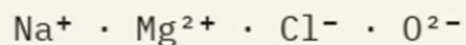
Al³⁺ N³⁻ O²⁻ S²⁻ Se²⁻ Te²⁻

A **monatomic ion** contains only one atom

- When an atom of a group of atoms loses or gains electrons, it becomes an **ion**.
- Cations** are formed when at least one electron is lost. Monatomic cations are formed by metals.
- Anions** are formed when at least one electron is gained. Monatomic anions are formed by nonmetals, except the noble gases.

Monatomic

A single atom carrying a charge.



Polyatomic

A bonded group of atoms carrying a single overall charge.



5.0 IONS

Do You Understand Ions?

How many protons and electrons are in ${}_{13}^{27}\text{Al}^{3+}$?

13 protons, 10 (13 – 3) electrons

How many protons and electrons are in ${}_{34}^{78}\text{Se}^{2-}$?

34 protons, 36 (34 + 2) electrons



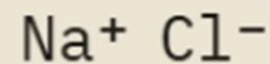
SECTION SIX

06 Chemical Formula

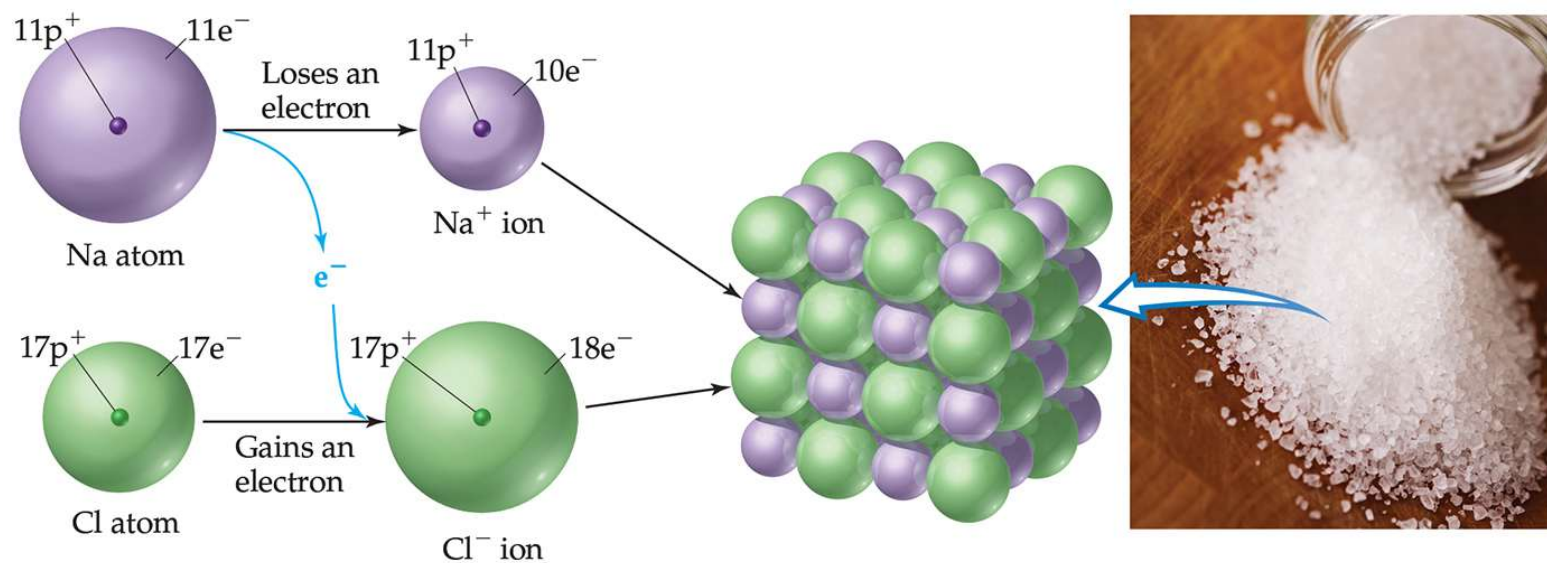
Every compound is electrically neutral — the charges must sum to zero.

6.0 CHEMICAL FORMULA

- A combination of cations and anions, generally formed between a metal and a nonmetal.
- Electrons transfer from metal to nonmetal.
- Opposite charges attract.
- Charges must sum to zero; only empirical formulas are written.



$$(+1) + (-1) = 0$$



**The ionic
compound
NaCl**



How is an ion formed?

- A. By either adding or subtracting protons from the atom
- B. By either adding or subtracting electrons from the atom
- C. By either adding or subtracting neutrons from the atom
- D. All of these are true
- E. Two of these are true

(POLYATOMIC ION)

A *polyatomic ion* contains more than one atom

Ions with -1 charge

perbromate	BrO_4^{-1}
bromate	BrO_3^{-1}
bromite	BrO_2^{-1}
hypobromite	BrO^{-1}
perchlorate	ClO_4^{-1}
chlorate	ClO_3^{-1}
chlorite	ClO_2^{-1}
hypochlorite	ClO^{-1}
periodate	IO_4^{-1}
iodate	IO_3^{-1}
iodite	IO_2^{-1}
hypoiodite	IO^{-1}

nitrate	NO_3^{-1}
nitrite	NO_2^{-1}
hydroxide	OH^{-1}
cyanide	CN^{-1}
thiocyanate	SCN^{-1}
acetate	$\text{C}_2\text{H}_3\text{O}_2^{-1}$
permanganate	MnO_4^{-1}
bicarbonate	HCO_3^{-1}

Ions with a -2 Charge

carbonate	CO_3^{-2}
phthalate	$\text{C}_8\text{H}_4\text{O}_4^{-2}$
sulfate	SO_4^{-2}

sulfite	SO_3^{-2}
chromate	CrO_4^{-2}
dichromate	$\text{Cr}_2\text{O}_7^{-2}$
oxalate	$\text{C}_2\text{O}_4^{-2}$
peroxide	O_2^{-2}

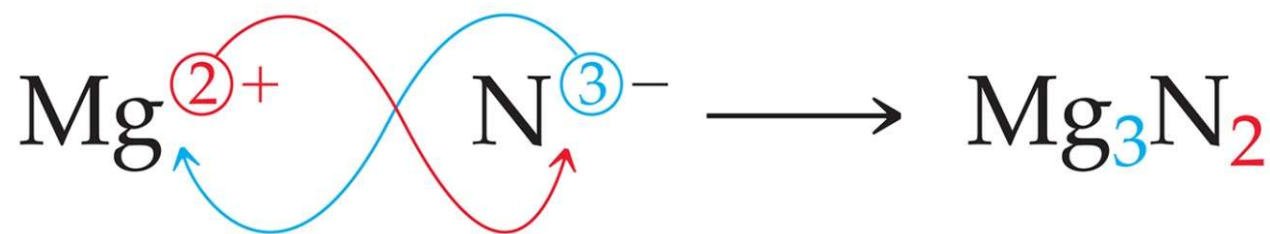
Ions with a -3 Charge

phosphate	PO_4^{-3}
phosphite	PO_3^{-3}
arsenate	AsO_4^{-3}

Ions with +1 charge

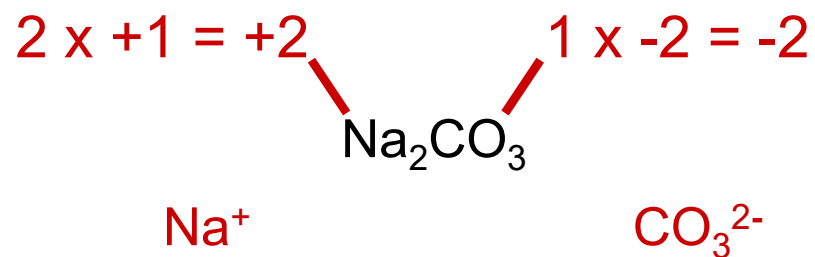
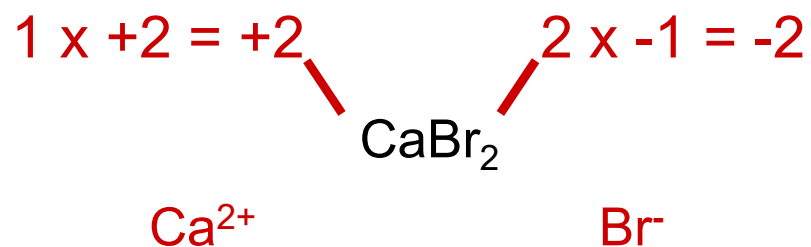
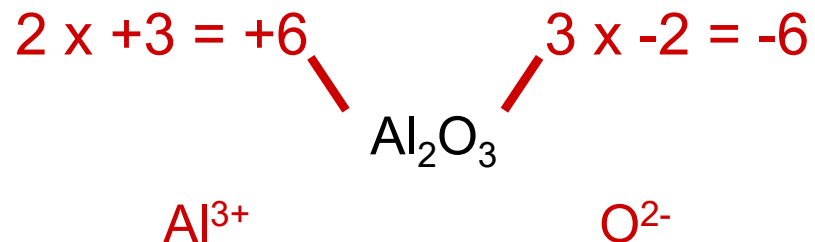
ammonium ion	NH_4^{+1}
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WRITING FORMULAS



- Because compounds are electrically neutral, one can determine the formula of a compound this way:
 - The charge on the cation becomes the subscript on the anion.
 - The charge on the anion becomes the subscript on the cation.
 - If these subscripts are not in the lowest whole-number ratio, divide them by the greatest common factor.

Formula of Ionic Compounds





SECTION SEVEN

07 Chemical Nomenclature

The systematic rules for naming compounds.

7.0 CHEMICAL NOMENCLATURE

What We Will Name

- 01 Ionic compounds
- 02 Transition-metal ionic compounds
- 03 Binary molecular compounds
- 04 Acids & oxoacids
- 05 Simple organic compounds — alkanes & alcohols

Periodic table of the elements

- Alkali metals
- Alkaline-earth metals
- Transition metals
- Other metals
- Other nonmetals
- Halogens
- Noble gases
- Rare-earth elements (21, 39, 57–71) and lanthanoid elements (57–71 only)
- Actinoid elements

Ionic bond = metal + Non metal

Covalent = Non metal + Non metal

Non Metal

Metal

Metallic Bond = metal + metal

group	1*	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	1 H																	2 He
2	3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
3	11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
6	55 Cs	56 Ba	57 La	72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
7	87 Fr	88 Ra	89 Ac	104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Mc	116 Lv	117 Ts	118 Og
lanthanoid series 6			58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu		
actinoid series 7			90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr		

7.0 CHEMICAL NOMENCLATURE

IONIC COMPOUNDS

- often a metal + nonmetal
- Cation is always named first and the anion second
- Monatomic cation takes its name from the name of the parent element
- Monatomic anion is named by taking the root of the element name and adding -ide
- anion (nonmetal), add “ide” to element name

Cation	Name	Anion	Name
H ⁺	Hydrogen	H ⁻	Hydride
Li ⁺	Lithium	F ⁻	Fluoride
Na ⁺	Sodium	Cl ⁻	Chloride
K ⁺	Potassium	Br ⁻	Bromide
Cs ⁺	Cesium	I ⁻	Iodide
Be ²⁺	Beryllium	O ²⁻	Oxide
Mg ²⁺	Magnesium	S ²⁻	Sulfide
Ca ²⁺	Calcium	N ³⁻	Nitride
Ba ²⁺	Barium	P ³⁻	Phosphide
Al ³⁺	Aluminum		

Name each binary compound

- CsF
 - AlCl₃
 - LiH
-
- CsF is cesium fluoride
 - AlCl₃ is aluminum chloride
 - LiH is lithium hydride

Common Monatomic Cations and Anions

7.0 CHEMICAL NOMENCLATURE

TABLE OF POLYATOMIC IONS					
acetate	CH_3COO^-	dihydrogen phosphate	H_2PO_4^-	oxalate	$\text{C}_2\text{O}_4^{2-}$
arsenate	AsO_4^{3-}	hydrogen carbonate	HCO_3^-	perchlorate	ClO_4^-
arsenite	AsO_3^{3-}	hydrogen oxalate	HC_2O_4^-	periodate	IO_4^-
benzoate	$\text{C}_6\text{H}_5\text{COO}^-$	hydrogen sulfate	HSO_4^-	permanganate	MnO_4^-
borate	BO_3^{3-}	hydrogen sulfide	HS^-	peroxide	O_2^{2-}
bromate	BrO_3^-	hydrogen sulfite	HSO_3^-	phosphate	PO_4^{3-}
carbonate	CO_3^{2-}	hydroxide	OH^-	pyrophosphate	$\text{P}_2\text{O}_7^{4-}$
chlorate	ClO_3^-	hypochlorite	ClO^-	sulfate	SO_4^{2-}
chloride	Cl^-	iodate	IO_3^-	sulfite	SO_3^{2-}
chlorite	ClO_2^-	monohydrogen phosphate	HPO_4^{2-}	thiocyanate	SCN^-
chromate	CrO_4^{2-}	nitrate	NO_3^-	thiosulfate	$\text{S}_2\text{O}_3^{2-}$
cyanate	CNO^-	nitrite	NO_2^-	POSITIVE POLYATOMIC IONS	
cyanide	CN^-	orthosilicate	SiO_4^{4-}	ammonium	NH_4^+
dichromate	$\text{Cr}_2\text{O}_7^{2-}$			hydronium	H_3O^+

PERIODIC TABLE OF IONS

KEY

atomic number → 26
ion charge → Fe^{3+}
symbol → Fe
ion name (IUPAC) → iron (III)
ion charge → Fe^{2+}
ion name (IUPAC) → iron (II)

1																	17	18			
1 H^+ hydrogen																	1 H^- hydride	2 He helium			
3 Li^+ lithium	4 Be^{2+} beryllium															13 B boron	14 C carbon	15 N^{3-} nitride	16 O^{2-} oxide	17 F^- fluoride	18 Ne neon
11 Na^+ sodium	12 Mg^{2+} magnesium															13 Al^{3+} aluminum	14 Si silicon	15 P^{3-} phosphide	16 S^{2-} sulfide	17 Cl^- chloride	18 Ar argon
19 K^+ potassium	20 Ca^{2+} calcium	21 Sc^{3+} scandium	22 Ti^{4+} titanium (IV)	23 V^{3+} vanadium(III)	24 Cr^{3+} chromium (III)	25 Mn^{2+} manganese(II)	26 Fe^{3+} iron (III)	27 Co^{2+} cobalt (II)	28 Ni^{2+} nickel (II)	29 Cu^{2+} copper (II)	30 Zn^{2+} zinc	31 Ga^{3+} gallium	32 Ge^{4+} germanium	33 As^{3-} arsenide	34 Se^{2-} selenide	35 Br^- bromide	36 Kr krypton				
37 Rb^+ rubidium	38 Sr^{2+} strontium	39 Y^{3+} yttrium	40 Zr^{4+} zirconium	41 Nb^{5+} niobium (V)	42 Mo^{6+} molybdenum	43 Tc^{7+} technetium	44 Ru^{3+} ruthenium(III)	45 Rh^{3+} rhodium	46 Pd^{2+} palladium(II)	47 Ag^+ silver	48 Cd^{2+} cadmium	49 In^{3+} indium	50 Sn^{4+} tin (IV)	51 Sb^{3+} antimony(III)	52 Te^{2-} telluride	53 I^- iodide	54 Xe xenon				
55 Cs^+ cesium	56 Ba^{2+} barium	57 La^{3+} lanthanum	72 Hf^{4+} hafnium	73 Ta^{5+} tantalum	74 W^{6+} tungsten	75 Re^{7+} rhenium	76 Os^{4+} osmium	77 Ir^{4+} iridium	78 Pt^{4+} platinum(IV)	79 Au^{3+} gold (III)	80 Hg^{2+} mercury (II)	81 Tl^+ thallium (I)	82 Pb^{2+} lead (II)	83 Bi^{3+} bismuth(III)	84 Po^{2+} polonium(II)	85 At^- astatide	86 Rn radon				
87 Fr^+ francium	88 Ra^{2+} radium	89 Ac^{3+} actinium																			
			58 Ce^{3+} cerium	59 Pr^{3+} praseodymium	60 Nd^{3+} neodymium	61 Pm^{3+} promethium	62 Sm^{3+} samarium(III)	63 Eu^{3+} europium (III)	64 Gd^{3+} gadolinium	65 Tb^{3+} terbium	66 Dy^{3+} dysprosium	67 Ho^{3+} holmium	68 Er^{3+} erbium	69 Tm^{3+} thulium	70 Yb^{3+} ytterbium(III)	71 Lu^{3+} lutetium					
			90 Th^{4+} thorium	91 Pa^{5+} protactinium(V)	92 U^{6+} uranium (VI)	93 Np^{5+} neptunium	94 Pu^{4+} plutonium(IV)	95 Am^{3+} americium(III)	96 Cm^{3+} curium	97 Bk^{3+} berkelium(III)	98 Cf^{3+} californium	99 Es^{3+} einsteinium	100 Fm^{3+} fermium	101 Md^{2+} mendelevium (II)	102 No^{2+} nobelium(II)	103 Lr^{3+} lawrencium					

7.0 CHEMICAL NOMENCLATURE

Transition metal ionic compounds

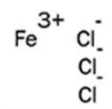
- indicate charge on metal with roman numerals

1 H ⁺ hydrogen	2 Li ⁺ lithium	3 Na ⁺ sodium	4 K ⁺ potassium	5 Rb ⁺ rubidium	6 Cs ⁺ cesium	7 Fr ⁺ francium	8 Be ²⁺ beryllium	9 Mg ²⁺ magnesium	10 Ca ²⁺ calcium	11 Sr ²⁺ strontium	12 Ba ²⁺ barium	13 Ra ²⁺ radium	14 Sc scandium	15 Y yttrium	16 La lanthanum	17 Ac ³⁺ actinium	18 He helium	19 Ne neon	20 Ar argon	21 Kr krypton	22 Xe xenon	23 Rn radon	24 Ti ⁴⁺ (titanium (IV))	25 V ³⁺ vanadium (III)	26 Cr ³⁺ chromium (III)	27 Mn ²⁺ manganese (II)	28 Fe ³⁺ iron (III)	29 Co ²⁺ cobalt (II)	30 Ni ²⁺ nickel (II)	31 Cu ²⁺ copper (II)	32 Zn ²⁺ zinc	33 Ga ³⁺ gallium	34 Ge ⁴⁺ germanium	35 As ³⁻ arsenide	36 Se ²⁻ selenide	37 Br ⁻ bromide	38 Kr krypton	39 Rb ⁺ rubidium	40 Sr ²⁺ strontium	41 Y ³⁺ yttrium	42 Zr ⁴⁺ zirconium	43 Nb ⁵⁺ niobium (V)	44 Mo ⁶⁺ molybdenum	45 Tc ⁷⁺ technetium	46 Ru ³⁺ ruthenium (III)	47 Rh ³⁺ rhodium	48 Pd ²⁺ palladium (II)	49 Ag ⁺ silver	50 Cd ²⁺ cadmium	51 In ³⁺ indium	52 Sn ⁴⁺ tin (IV)	53 Sb ³⁺ antimony (III)	54 Te ²⁻ telluride	55 I ⁻ iodide	56 Xe xenon	57 Cs ⁺ cesium	58 Ba ²⁺ barium	59 La lanthanum	60 Hf ⁴⁺ hafnium	61 Ta ⁵⁺ tantalum	62 W ⁶⁺ tungsten	63 Re ⁷⁺ rhenium	64 Os ⁴⁺ osmium	65 Ir ⁴⁺ iridium	66 Pt ⁴⁺ platinum (IV)	67 Au ³⁺ gold (III)	68 Hg ²⁺ mercury (II)	69 Tl ³⁺ thallium (III)	70 Pb ²⁺ lead (II)	71 Bi ³⁺ bismuth (III)	72 Po ²⁺ polonium (II)	73 At ⁻ astatine	74 Rn radon	75 Fr ⁺ francium	76 Ra ²⁺ radium	77 Ac ³⁺ actinium	78 Ce ³⁺ cerium	79 Pr ³⁺ praseodymium	80 Nd ³⁺ neodymium	81 Pm ³⁺ promethium	82 Sm ³⁺ samarium (III)	83 Eu ³⁺ europium (III)	84 Gd ³⁺ gadolinium	85 Tb ³⁺ terbium	86 Dy ³⁺ dysprosium	87 Ho ³⁺ holmium	88 Er ³⁺ erbium	89 Tm ³⁺ thulium	90 Yb ³⁺ ytterbium (III)	91 Lu ³⁺ lutetium	92 Th ⁴⁺ thorium	93 Pa ⁵⁺ protactinium (V)	94 U ⁶⁺ uranium (VI)	95 Np ⁵⁺ neptunium	96 Pu ⁴⁺ plutonium (IV)	97 Am ³⁺ americium (III)	98 Cm ³⁺ curium	99 Bk ³⁺ berkelium (III)	100 Cf ³⁺ californium	101 Es ³⁺ einsteinium	102 Fm ³⁺ fermium	103 Md ³⁺ mendelevium (III)	104 No ³⁺ nobelium (III)	105 Lr ³⁺ lawrencium
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Zinc is always Zn²⁺ ; silver is always Ag⁺

			Systematic Name
FeCl ₂	2Cl ⁻	-2 so Fe is +2	Iron (II) Chloride
FeCl ₃	3Cl ⁻	-3 so Fe is +3	Iron (III) Chloride
Cr ₂ S ₃	3S ²⁻	-6 so Cr is +3 (6/2)	Chromium (III) sulfide

Ex:



+3 -3
Iron(III) chloride

POLYATOMIC IONS

Negative Ions

Fluoride	F ⁻
Chloride	Cl ⁻
Bromide	Br ⁻
Iodide	I ⁻
Oxide	O ²⁻
Sulfide	S ²⁻
Nitride	N ³⁻
Phosphide	P ³⁻

Carbonate	CO ₃ ²⁻
Chromite	CrO ₂ ⁻
Hypochlorite	ClO ⁻
Nitrate	NO ₃ ⁻
Nitrite	NO ₂ ⁻
Permanganate	MnO ₄ ⁻
Phosphate	PO ₄ ³⁻
Phosphite	PO ₃ ³⁻
Sulfate	SO ₄ ²⁻
Sulfite	SO ₃ ²⁻

1 = I
2 = II
3 = III
4 = IV
5 = V
6 = VI
7 = VII
8 = VIII
9 = IX
10 = X

11 = XI
12 = XII
13 = XIII
14 = XIV
15 = XV
16 = XVI
17 = XVII
18 = XVIII
19 = XIX
20 = XX

7.0 CHEMICAL NOMENCLATURE

Transition metal ionic compounds

- indicate charge on metal with roman numerals

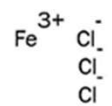
1 H ⁺ hydrogen	2 He ²⁺ helium										
3 Li ⁺ lithium	4 Be ²⁺ beryllium	21 Sc scandium	22 Ti titanium	23 V vanadium	24 Cr chromium	25 Mn manganese	26 Fe iron	27 Co cobalt	28 Ni nickel	29 Cu copper	30 Zn zinc
				2+	2+	2+	2+	2+	2+	1+	
		3+	3+	3+	3+	3+	3+	3+	3+	2+	2+
			4+	4+	4+	4+					
			5+								
					6+	6+	6+				
					7+						

90 Th ⁴⁺ thorium	91 Pa ⁵⁺ protactinium(V)	92 U ⁶⁺ uranium(VI)	93 Np ⁵⁺ neptunium(V)	94 Pu ⁴⁺ plutonium(IV)	95 Am ³⁺ americium(III)	96 Cm ³⁺ curium(III)	97 Bk ³⁺ berkelium(III)	98 Cf ³⁺ californium(III)	99 Es ³⁺ einsteinium(III)	100 Fm ³⁺ fermium(III)	101 Md ²⁺ mendelevium(II)	102 No ²⁺ nobelium(II)	103 Lr ³⁺ lawrencium(III)
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Zinc is always Zn²⁺ ; silver is always Ag⁺

			Systematic Name
FeCl ₂	2Cl ⁻	-2 so Fe is +2	Iron (II) Chloride
FeCl ₃	3Cl ⁻	-3 so Fe is +3	Iron (III) Chloride
Cr ₂ S ₃	3S ²⁻	-6 so Cr is +3 (6/2)	Chromium (III) sulfide

Ex:



+3 -3
Iron(III) chloride

Negative Ions

Fluoride	F ⁻
Chloride	Cl ⁻
Bromide	Br ⁻
Iodide	I ⁻
Oxide	O ²⁻
Sulfide	S ²⁻
Nitride	N ³⁻
Phosphide	P ³⁻

POLYATOMIC IONS

Carbonate	CO ₃ ²⁻
Chromite	CrO ₂ ⁻
Hypochlorite	ClO ⁻
Nitrate	NO ₃ ⁻
Nitrite	NO ₂ ⁻
Permanganate	MnO ₄ ⁻
Phosphate	PO ₄ ³⁻
Phosphite	PO ₃ ³⁻
Sulfate	SO ₄ ²⁻
Sulfite	SO ₃ ²⁻

1 = I
2 = II
3 = III
4 = IV
5 = V
6 = VI
7 = VII
8 = VIII
9 = IX
10 = X

11 = XI
12 = XII
13 = XIII
14 = XIV
15 = XV
16 = XVI
17 = XVII
18 = XVIII
19 = XIX
20 = XX

More Example:

1. Chemical formulae of ionic compounds:

a. iron (III) chloride



b. nickel (II) oxide



c. chromium (VI) oxide



d. silver (II) chloride



e. vanadium (II) bromide



f. zinc (I) sulphate



g. iron (III) hydroxide



h. manganese (II) oxide



i. manganese (III) oxide



j. copper (II) sulphate



k. mercury (II) chloride



2. Write a formula for the following pairs of ions and then write a name for them. Remember, only compounds of transition metals need the Roman numerals. The first one is done for you.

a. Cu^{2+} and Cl^-	CuCl_2	copper (II) chloride
b. Ti^{2+} and O^{2-}	TiO	titanium (II) oxide
c. Ca^{2+} and S^{2-}	CaS	calcium ^{sulfide} sulphate
d. Na ion and N^{3-}	Na_3N	sodium nitride
e. Mg ion and OH^-	$\text{Mg}(\text{OH})_2$	magnesium hydroxide
f. Zn^{2+} and (NO_3^-)	$\text{Zn}(\text{NO}_3)_2$	zinc (II) nitrate
g. K^+ and F^-	KF	potassium fluoride

7.0 CHEMICAL NOMENCLATURE

H_2O Group (left to right!)

TABLE OF MONOATOMIC IONS

acetate	CH_3COO^-
arsenate	AsO_4^{3-}
arsenite	AsO_3^{3-}
benzoate	$C_6H_5COO^-$
borate	BO_3^{3-}
bromate	BrO_3^-
carbonate	CO_3^{2-}
chlorate	ClO_3^-
chloride	Cl^-
chlorite	ClO_2^-
chromate	CrO_4^{2-}
cyanate	CNO^-
cyanide	CN^-
dichromate	$Cr_2O_7^{2-}$

TABLE OF POLYATOMIC IONS

dihydrogen phosphate	$H_2PO_4^-$
hydrogen carbonate	HCO_3^-
hydrogen oxalate	$HC_2O_4^-$
hydrogen sulfate	HSO_4^-
hydrogen sulfide	HS^-
hydrogen sulfite	HSO_3^-
hydroxide	OH^-
hypochlorite	ClO^-
iodate	IO_3^-
monohydrogen phosphate	HPO_4^{2-}
nitrate	NO_3^-
nitrite	NO_2^-
orthosilicate	SiO_4^{4-}

PERIODIC TABLE OF IONS

KEY:

atomic number	26	ion charge	Fe^{3+}
		ion name (IUPAC)	iron (III)
symbol	Fe^{2+}		iron (II)

SO₂ (greater period first!)

PERIODIC TABLE OF ELEMENTS

1	2																	17	18
1	2																	17	18
1	2																	17	18
3	4																	17	18
3	4																	17	18
11	12																	17	18
11	12																	17	18
19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36		
19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36		
37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54		
37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54		
55	56	57	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86		
55	56	57	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86		
87	88	89															117	118	
87	88	89															117	118	

Naming Covalent Compounds

NOMENCLATURE OF BINARY MOLECULAR COMPOUNDS (INORGANIC COMPOUNDS)

**TABLE 2.6 Prefixes Used in
Naming Binary Compounds
Formed between Nonmetals**

Prefix	Meaning
<i>mono-</i>	1
<i>di-</i>	2
<i>tri-</i>	3
<i>tetra-</i>	4
<i>penta-</i>	5
<i>hexa-</i>	6
<i>hepta-</i>	7
<i>octa-</i>	8
<i>nona-</i>	9
<i>deca-</i>	10

- THE NAME OF THE ELEMENT FARTHER TO THE LEFT IN THE PERIODIC TABLE (CLOSER TO THE METALS) OR LOWER IN THE SAME GROUP IS USUALLY WRITTEN FIRST.
- A PREFIX IS USED TO DENOTE THE NUMBER OF ATOMS OF EACH ELEMENT IN THE COMPOUND (**MONO-** IS NOT USED ON THE FIRST ELEMENT LISTED, HOWEVER).

Naming Covalent Compounds

Rule 1: Element with lower group number is first

Rule 2: If in same group the greater period is first

Rule 3: Second element is an anion (-ide)

Rule 4: Greek prefixes determine number*

* no mono on first, monoxide = monoxide



Prefix	Number Indicated
<i>mono-</i>	1
<i>di-</i>	2
<i>tri-</i>	3
<i>tetra-</i>	4
<i>penta-</i>	5
<i>hexa-</i>	6
<i>hepta-</i>	7
<i>octa-</i>	8
<i>nona-</i>	9
<i>deca-</i>	10

Writing Covalent Formula

- dihydrogen monoxide
- dinitrogen trisulfide
- silicon dihydride
- sulfur hexafluoride

INTERACTIVE EXAMPLE

1. NAME EACH OF THE FOLLOWING COMPOUNDS:



2. FROM THE FOLLOWING SYSTEMATIC NAMES, WRITE THE FORMULA FOR EACH COMPOUND:

a. SULFUR HEXAFLUORIDE

b. SULFUR TRIOXIDE

c. CARBON DIOXIDE

The “-ide” Nomenclature of Some Common Monoatomic Anions According to Their Positions in the Periodic Table

Group 4A	Group 5A	Group 6A	Group 7A
C carbide (C^{4-})*	N nitride (N^{3-})	O oxide (O^{2-})	F fluoride (F^-)
Si silicide (Si^{4-})	P phosphide (P^{3-})	S sulfide (S^{2-})	Cl chloride (Cl^-)
		Se selenide (Se^{2-})	Br bromide (Br^-)
		Te telluride (Te^{2-})	I iodide (I^-)

NOMENCLATURE OF BINARY COMPOUNDS

- THE ENDING ON THE SECOND ELEMENT IS CHANGED TO **-IDE**.
 - CO_2 : CARBON DIOXIDE
 - CCl_4 : CARBON TETRACHLORIDE
- If the prefix ends with *a* or *o* and the name of the element begins with a vowel, the two successive vowels are often elided into one.
 - N_2O_5 : DINITROGEN PENTOXIDE
 - CO : CARBON MONOXIDE

Names and Formulas of Some Common Inorganic Cations & Anions

Cation	Anion
aluminum (Al^{3+})	bromide (Br^-)
ammonium (NH_4^+)	carbonate (CO_3^{2-})
barium (Ba^{2+})	chlorate (ClO_3^-)
cadmium (Cd^{2+})	chloride (Cl^-)
calcium (Ca^{2+})	chromate (CrO_4^{2-})
cesium (Cs^+)	cyanide (CN^-)
chromium(III) or chromic (Cr^{3+})	dichromate ($\text{Cr}_2\text{O}_7^{2-}$)
cobalt(II) or cobaltous (Co^{2+})	dihydrogen phosphate (H_2PO_4^-)
copper(I) or cuprous (Cu^+)	fluoride (F^-)
copper(II) or cupric (Cu^{2+})	hydride (H^-)
hydrogen (H^+)	hydrogen carbonate or bicarbonate (HCO_3^-)
iron(II) or ferrous (Fe^{2+})	hydrogen phosphate (HPO_4^{2-})
iron(III) or ferric (Fe^{3+})	hydrogen sulfate or bisulfate (HSO_4^-)
lead(II) or plumbous (Pb^{2+})	hydroxide (OH^-)
lithium (Li^+)	iodide (I^-)
magnesium (Mg^{2+})	nitrate (NO_3^-)
manganese(II) or manganous (Mn^{2+})	nitride (N^{3-})
mercury(I) or mercurous (Hg_2^{2+})*	nitrite (NO_2^-)
mercury(II) or mercuric (Hg^{2+})	oxide (O^{2-})
potassium (K^+)	permanganate (MnO_4^-)
rubidium (Rb^+)	peroxide (O_2^{2-})
silver (Ag^+)	phosphate (PO_4^{3-})
sodium (Na^+)	sulfate (SO_4^{2-})
strontium (Sr^{2+})	sulfide (S^{2-})
tin(II) or stannous (Sn^{2+})	sulfite (SO_3^{2-})
zinc (Zn^{2+})	thiocyanate (SCN^-)

Some Polyatomic Ions

NH_4^+	ammonium	SO_4^{2-}	sulfate
CO_3^{2-}	carbonate	SO_3^{2-}	sulfite
HCO_3^-	bicarbonate	NO_3^-	nitrate
ClO_3^-	chlorate	NO_2^-	nitrite
$\text{Cr}_2\text{O}_7^{2-}$	dichromate	SCN^-	thiocyanate
CrO_4^{2-}	chromate	OH^-	hydroxide

PATTERNS IN OXYANION NOMENCLATURE

An oxyanion, or oxoanion, is an ion with the generic formula $A_xO_y^{z-}$ (where A represents a chemical element and O represents an oxygen atom).

- When there are two oxyanions involving the same element
 - The one with fewer /with smaller oxygens ends in **-ITE**.
 - The one with more oxygens/large number ends in **-ATE**.
- NO_2^- : NITRITE; NO_3^- : NITRATE
- SO_3^{2-} : SULFITE; SO_4^{2-} : SULFATE

		TABLE OF POLYATOMIC IONS			
acetate	CH_3COO^-	dihydrogen phosphate	H_2PO_4^-	oxalate	$\text{C}_2\text{O}_4^{2-}$
arsenate	AsO_4^{3-}	hydrogen carbonate	HCO_3^-	perchlorate	ClO_4^-
arsenite	AsO_3^{3-}	hydrogen oxalate	HC_2O_4^-	periodate	IO_4^-
benzoate	$\text{C}_6\text{H}_5\text{COO}^-$	hydrogen sulfate	HSO_4^-	permanganate	MnO_4^-
borate	BO_3^{3-}	hydrogen sulfide	HS^-	peroxide	O_2^{2-}
bromate	BrO_3^-	hydrogen sulfite	HSO_3^-	phosphate	PO_4^{3-}
carbonate	CO_3^{2-}	hydroxide	OH^-	pyrophosphate	$\text{P}_2\text{O}_7^{4-}$
chlorate	ClO_3^-	hypochlorite	ClO^-	sulfate	SO_4^{2-}
chloride	Cl^-	iodate	IO_3^-	sulfite	SO_3^{2-}
chlorite	ClO_2^-	monohydrogen phosphate	HPO_4^{2-}	thiocyanate	SCN^-
chromate	CrO_4^{2-}	nitrate	NO_3^-	thiosulfate	$\text{S}_2\text{O}_3^{2-}$
cyanate	CNO^-	nitrite	NO_2^-	POSITIVE POLYATOMIC IONS	
cyanide	CN^-	orthosilicate	SiO_4^{4-}	ammonium	NH_4^+
dichromate	$\text{Cr}_2\text{O}_7^{2-}$			hydronium	H_3O^+

POLYATOMIC IONS

- WHEN MORE THAN TWO OXYANIONS MAKE UP A SERIES:
 - USE THE PREFIX HYPO- (LESS THAN) TO NAME MEMBERS OF THE SERIES WITH THE FEWEST O ATOMS
 - USE THE PREFIX PER- (MORE THAN) TO NAME MEMBERS OF THE SERIES WITH THE MOST O ATOMS

Ion	Name	Ion	Name
Hg_2^{2+}	Mercury(I)	CO_3^{2-}	Carbonate
NH_4^+	Ammonium	HCO_3^-	Hydrogen carbonate (bicarbonate is a widely used common name)
NO_2^-	Nitrite	ClO^- or OCl^-	Hypochlorite
NO_3^-	Nitrate	ClO_2^-	Chlorite
SO_3^{2-}	Sulfite	ClO_3^-	Chlorate
SO_4^{2-}	Sulfate	ClO_4^-	Perchlorate
HSO_4^-	Hydrogen sulfate (bisulfate is a widely used common name)	$\text{C}_2\text{H}_3\text{O}_2$	Acetate
OH^-	Hydroxide	MnO_4^-	Permanganate
CN^-	Cyanide	$\text{Cr}_2\text{O}_7^{2-}$	Dichromate
PO_4^{3-}	Phosphate	CrO_4^{2-}	Chromate
HPO_4^{2-}	Hydrogen phosphate	O_2^{2-}	Peroxide
H_2PO_4^-	Dihydrogen phosphate	$\text{C}_2\text{O}_4^{2-}$	Oxalate
NCS^- or SCN^-	Thiocyanate	$\text{S}_2\text{O}_3^{2-}$	Thiosulfate

PATTERNS IN OXYANION NOMENCLATURE

Elements that can form more than one **oxyanion** (**polyatomic ions that contain oxygen**), each containing a different number of oxygen atoms.



In a family of oxyanions, **the charge remains the same**; only the number of oxygen atom varies

the **most common of an element's oxyanions** has a name with the form **(root)ate**.

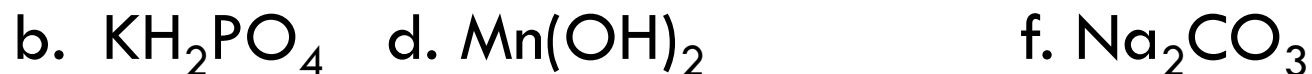
Relationship	General Name	Example Name	Example formula
One more oxygen atom than (root)ate	per (root)ate	perchlorate	ClO_4^-
Most Common	(root)ate	chlorate	ClO_3^-
One less oxygen than (root)ate	(root) ite	chlorite	ClO_2^-
Two less oxygen than (root)ate	hypo (root) ite	hypochlorite	ClO^-

	Rule	Example
most	per + "root" + ate	perchlorate ClO_4^-
more	"root" + ate	chlorate ClO_3^-
less	"root" + ite	chlorite ClO_2^-
least	hypo + "root" + ite	hypochlorite ClO^-

ClO^-	hypo- · -ite	hypochlorite
ClO_2^-	· -ite	chlorite
ClO_3^-	· -ate	chlorate
ClO_4^-	per- · -ate	perchlorate

INTERACTIVE EXAMPLE -NAMING COMPOUNDS CONTAINING POLYATOMIC IONS

1. Give the systematic name for each of the following compounds:



2. Given the following systematic names, write the formula for each compound:

a. Sodium hydrogen carbonate d. Sodium selenate

b. Cesium perchlorate e. Potassium bromate

c. Sodium hypochlorite

ACID NOMENCLATURE

Anion		Acid
____ide (chloride, Cl^-)	add H^+	hydro____ic acid (hydrochloric acid, HCl)
____ate (chlorate, ClO_3^-) (perchlorate, ClO_4^-)	add H^+	____ic acid (chloric acid, HClO_3) (perchloric acid, HClO_4)
____ite (chlorite, ClO_2^-) (hypochlorite, ClO^-)	add H^+	____ous acid (chlorous acid, HClO_2) (hypochlorous acid, HClO)

- If the anion in the acid ends in **-ide**, change the ending to **-ic acid** and add the prefix **hydro-**.

- HCl : HYDROCHLORIC ACID
- HBr : HYDROBROMIC ACID
- HI : HYDROIODIC ACID

- If the anion ends in **-ate**, change the ending to **-ic acid**.
 - HClO_3 : chloric acid
 - HClO_4 : perchloric acid
- If the anion ends in **-ite**, change the ending to **-ous acid**.
 - HClO : hypochlorous acid
 - HClO_2 : chlorous acid

	Anion Ending	Acid Name
No Oxygen →	-ide	hydro-(stem)-ic acid
w/Oxygen ↗	-ate	(stem)-ic acid
	-ite	(stem)-ous acid

My **ride** has **hydrolics**
 I **ate** something **icky**
Sprite is delici**ous**

Names of Oxoacids and Oxoanions that contain Chlorine

Acid	Corresponding Anion
HClO_4 (perchloric acid)	ClO_4^- (perchlorate)
HClO_3 (chloric acid)	ClO_3^- (chlorate)
HClO_2 (chlorous acid)	ClO_2^- (chlorite)
HClO (hypochlorous acid)	ClO^- (hypochlorite)

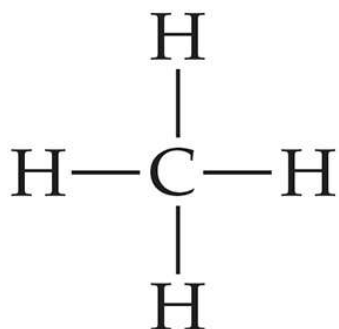
Names of Acids* That Do Not Contain Oxygen

Acid	Name
HF	Hydrofluoric acid
HCl	Hydrochloric acid
HBr	Hydrobromic acid
HI	Hydroiodic acid
HCN	Hydrocyanic acid
H_2S	Hydrosulfuric acid

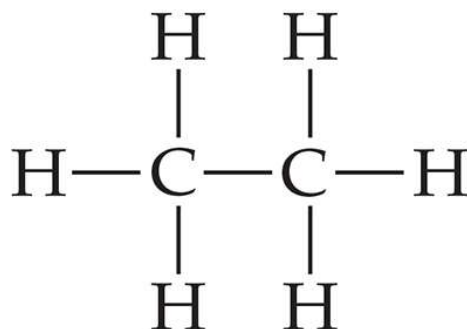
Names of Some Oxygen-Containing Acids

Acid	Name
HNO_3	Nitric acid
HNO_2	Nitrous acid
H_2SO_4	Sulfuric acid
H_2SO_3	Sulfurous acid
H_3PO_4	Phosphoric acid
$\text{HC}_2\text{H}_3\text{O}_2$	Acetic acid

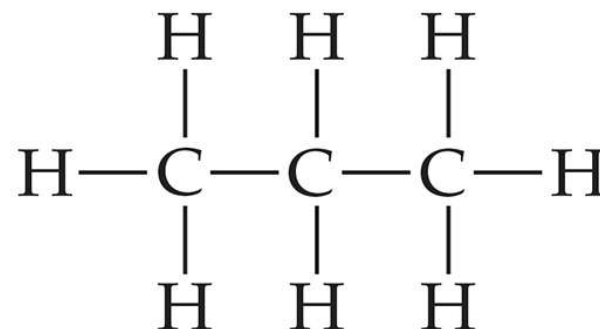
NOMENCLATURE OF ORGANIC COMPOUNDS: ALKANES



Methane



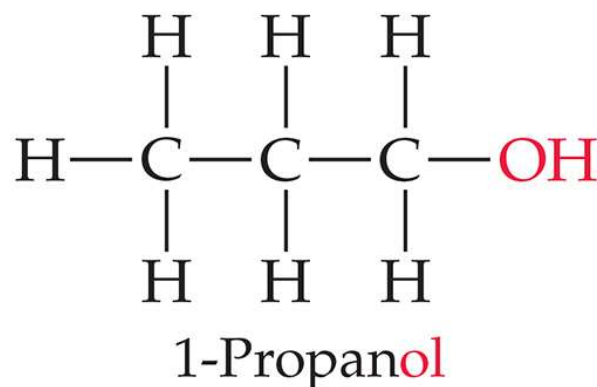
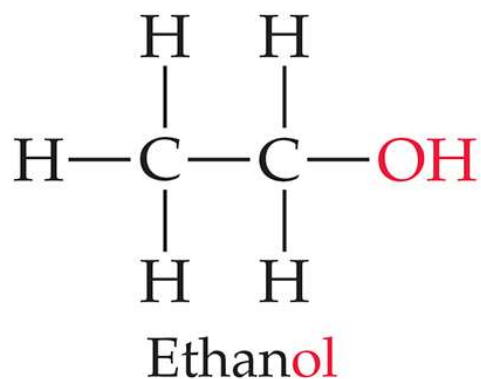
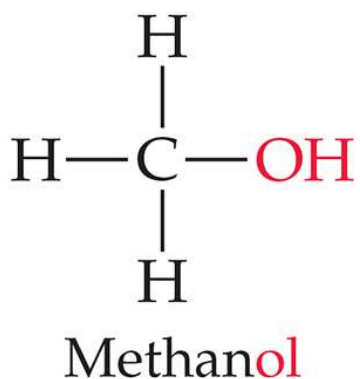
Ethane



Propane

- **Organic chemistry** is the study of carbon.
- Organic chemistry has its own system of nomenclature.
- The simplest hydrocarbons (compounds containing only carbon and hydrogen) are **alkanes**.
- The first part of the names just listed correspond to the number of carbons (*meth-* = 1, *eth-* = 2, *prop-* = 3, etc.).
- It is followed by **-ane**.

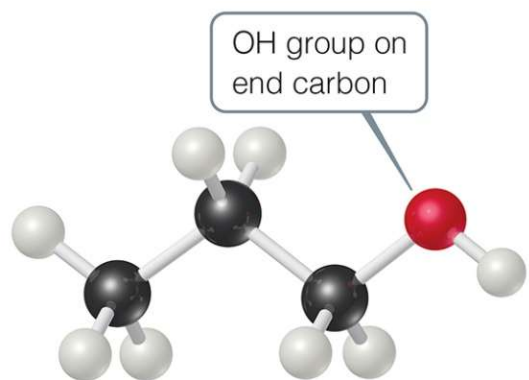
NOMENCLATURE OF ORGANIC COMPOUNDS: ALCOHOLS



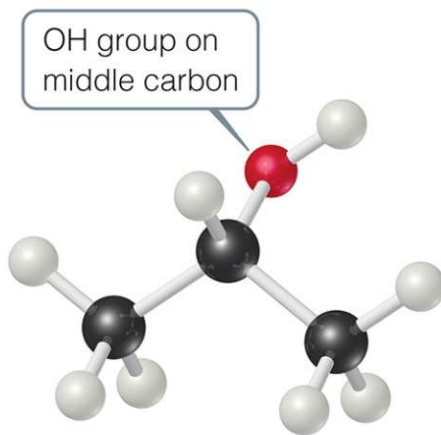
- When a hydrogen in an alkane is replaced with something else (a **functional group**, like $-\text{OH}$ in the compounds above), the name is derived from the name of the alkane.
- The ending denotes the type of compound.
 - An **alcohol** ends in **-ol**.

NOMENCLATURE OF ORGANIC COMPOUNDS: ALCOHOLS

- When two or more molecules have the same chemical formula, but different structures, they are called **isomers**.
- 1-propanol and 2-propanol have the oxygen atom connected to different carbon atoms, but both have the formula $\text{C}_3\text{H}_8\text{O}$.



1-Propanol



2-Propanol



SECTION EIGHT

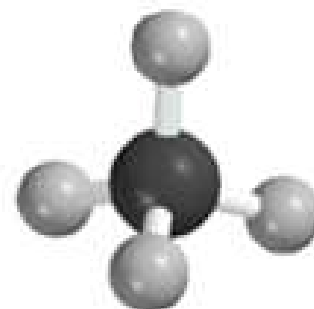
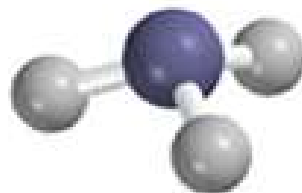
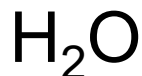
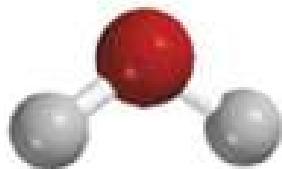
08 Molecular Compounds

Atoms held together by chemical bonds — almost always nonmetals only.

8.0 MOLECULAR COMPOUNDS

Molecular compounds are composed of molecules and almost always contain only nonmetals.

A **molecule** consists of two or more atoms in a definite arrangement held together by chemical bonds.



A **diatomic molecule** contains only two atoms



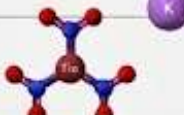
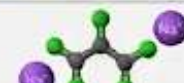
A **polyatomic molecule** contains more than two atoms



8.0 MOLECULAR COMPOUNDS

'CONSOLIDATED GUIDE TO CHEMICAL NOMENCLATURE & RELATIVE MASS'

1. FORMULA TO SYSTEMATIC NAME

a. Na_2SO_4	→ Sodium sulfate	
b. KH_2PO_4	→ Potassium dihydrogen phosphate	
c. $\text{Fe}(\text{NO}_3)_3$	→ Iron(III) nitrate	
d. $\text{Mn}(\text{OH})_2$	→ Manganese(II) hydroxide	
e. Na_2SO_3	→ Sodium sulfite	
f. Na_2CO_3	→ Sodium carbonate	

2. NAME TO FORMULA

a. Sodium hydrogen carbonate	→ NaHCO_3	Often called sodium bicarbonate 
b. Cesium perchlorate	→ CsClO_4	
c. Sodium hypochlorite	→ NaOCl	
d. Sodium selenate	→ Na_2SeO_4	Follows sulfur group pattern, similar to Na_2SO_4 
e. Potassium bromate	→ KBrO_3	

3. MOLECULAR COMPOUNDS: FORMULAS & MODELS

	Formula	Structural Formula	Ball-and-stick	Space-filling
Hydrogen	H_2	$\text{H}-\text{H}$		
Water	H_2O	$\text{H}-\text{O}-\text{H}$		
Ammonia	NH_3	$\text{H}-\text{N}-\text{H}$ $\quad $ $\quad \text{H}$		
Methane	CH_4	$\text{H}-\text{C}-\text{H}$ $\quad $ $\quad \text{H}$		

4. Advanced Molecular Nomenclature & Relative Mass

Naming Examples & Properties

HI	hydrogen iodide	
NF_3	nitrogen trifluoride	
SO_2	sulfur dioxide	
N_2Cl_4	dinitrogen tetrachloride TOXIC! ☠️	☠️
NO_2	nitrogen dioxide	
N_2O	dinitrogen monoxide Laughing Gas 😄	😄

Relative Mass Principles

Relative Atomic Mass (RAM)

The relative atomic mass (A_r) of an element is the average of the masses of the isotopes in a naturally occurring sample of the element relative to the mass of 1/12 of an atom of carbon-12.

$$\text{RAM} = \frac{\text{Average masses of one atom of an element}}{(1/12 \text{ of atom of carbon } 12)}$$

Relative Molecular Mass (RMM)

The relative molecular mass (M_r) of a compound is the mass of a molecule of that compound relative to the mass of 1/12 of an atom of carbon-12.

$$\text{RMM} = \frac{\text{Average masses of one molecule of a compound}}{(1/12 \text{ of atom of carbon } 12)}$$



SECTION NINE

09 Relative Mass

Masses compared against carbon-12.

9.0 RELATIVE MASS | Why Carbon-12?

The International Standard

- Instituted in 1961
- Standard chosen: ^{12}C (carbon-12)
- ^{12}C is assigned a mass of exactly 12 atomic mass units (u)
- All other atomic masses are measured relative to this standard

Why ^{12}C and not ^1H ?

Carbon-12 is easy to obtain in pure form, chemically stable, and gives whole-number masses for most common isotopes → making it the most practical universal reference.

Unified Atomic Mass Unit (u)

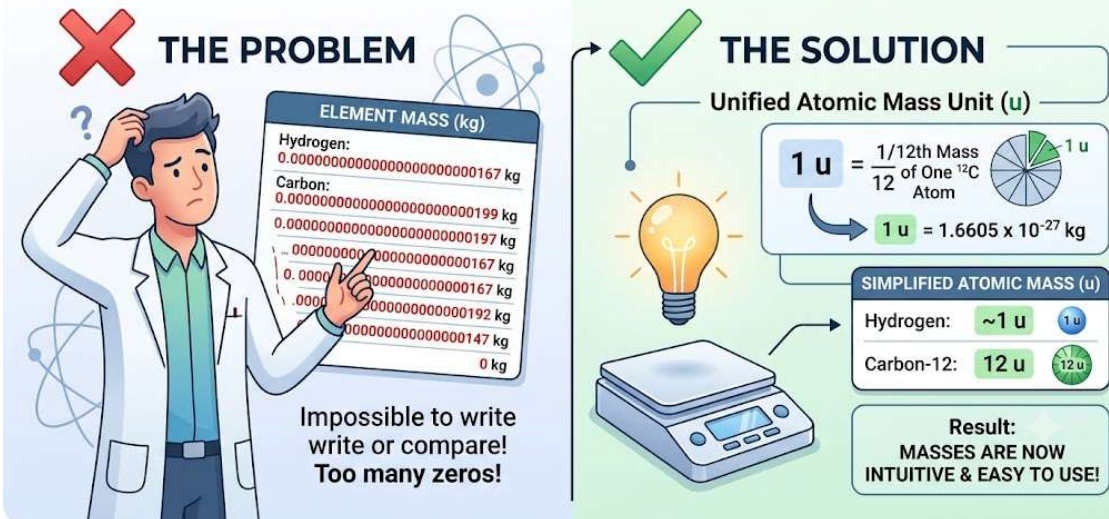
Mass of ^{12}C $\rightarrow 12 \text{ u}$

This is the reference unit

$1/12$ of $^{12}\text{C} = 1 \text{ u}$

1 u = 1.6605 x 10⁻²⁷ kg

MAKING ATOMIC MASS EASY TO READ.



9.0 RELATIVE MASS | Relative Isotopic Mass

Relative Isotopic Mass

The mass of one atom of an isotope compared to 1/12 the mass of one atom of ^{12}C .

For a pure isotope, the relative isotopic mass equals its mass number.

35
Cl

Relative isotopic mass = 35

37
Cl

Relative isotopic mass = 37

Worked Example: Chlorine (Cl)

^{35}Cl 75.77%

^{37}Cl 24.23%

$$\begin{aligned}\text{Ar}(\text{Cl}) &= \frac{(75.77 \times 35) + (24.23 \times 37)}{100} \\ &= \frac{(2651.95 + 896.51)}{100} \\ &= 3548.46 / 100\end{aligned}$$

$$\text{Ar}(\text{Cl}) = 35.5$$

Ar is the weighted average of all relative isotopic masses

9.0 RELATIVE MASS

Relative Atomic Mass (A_r)

Definition

The weighted mean mass of one atom of an element compared to $1/12$ the mass of one ^{12}C atom.

A_r = sum of (isotope mass x fractional abundance)

Worked Example: Chlorine (Cl)

Isotopes: ^{35}Cl (75.77%) and ^{37}Cl (24.23%)

$A_r = (35 \times 0.7577) + (37 \times 0.2423)$

$A_r(\text{Cl}) = 35.5$

A_r is dimensionless -- no units

Relative Molecular / Formula Mass (M_r)

Key Distinction

Covalent compound → Relative Molecular Mass (M_r)

Ionic compound → Relative Formula Mass (M_r)

M_r = sum of all A_r values in one molecule / formula unit

Worked Examples

H_2O (molecular): $M_r = 2(1) + 16 = \mathbf{18}$

NaCl (ionic): $M_r = 23 + 35.5 = \mathbf{58.5}$

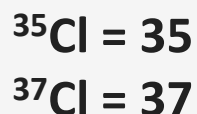
M_r is dimensionless -- no units

9.0 RELATIVE MASS | Summary

Relative Isotopic Mass

Mass of one atom of a specific isotope relative to $1/12$ mass of ^{12}C

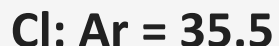
Example



One specific isotope

Relative Atomic Mass (A_r)

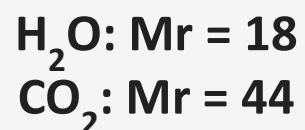
Weighted average mass of one atom of an element relative to $1/12$ mass of ^{12}C



An element (mixture of isotopes)

Relative Molecular Mass (M_r)

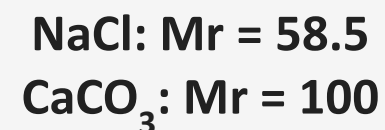
Mass of one molecule of a covalent compound relative to $1/12$ mass of ^{12}C



Covalent / molecular compounds

Relative Formula Mass (M_r)

Mass of one formula unit of an ionic compound relative to $1/12$ mass of ^{12}C



Ionic compounds

All four quantities are dimensionless ratios -- no units

9.0 RELATIVE MASS | Ar vs Atomic Mass vs Molar Mass

The same number -- three different questions. Know which one is being asked.

Relative Atomic Mass

Ar

No units (dimensionless)

A pure ratio comparing the average mass of one atom to 1/12 mass of ^{12}C .

Ar (Cl) = 35.5

How heavy relative to ^{12}C ?

Atomic Mass

35.5 u

Unit: u (unified atomic mass unit)

The actual mass of one atom expressed in unified atomic mass units (u).

Cl atom = 35.5 u

What is the real mass of one atom?

Molar Mass

35.5 g mol⁻¹

Unit: g mol⁻¹

The mass of one mole (6.02×10^{23} atoms) of the element.

Cl: M = 35.5 g mol⁻¹

Mass of one mole of atoms?

Ar and Mr are dimensionless. Atomic mass uses u. Molar mass uses g mol⁻¹.
Never mix them.



SECTION TEN

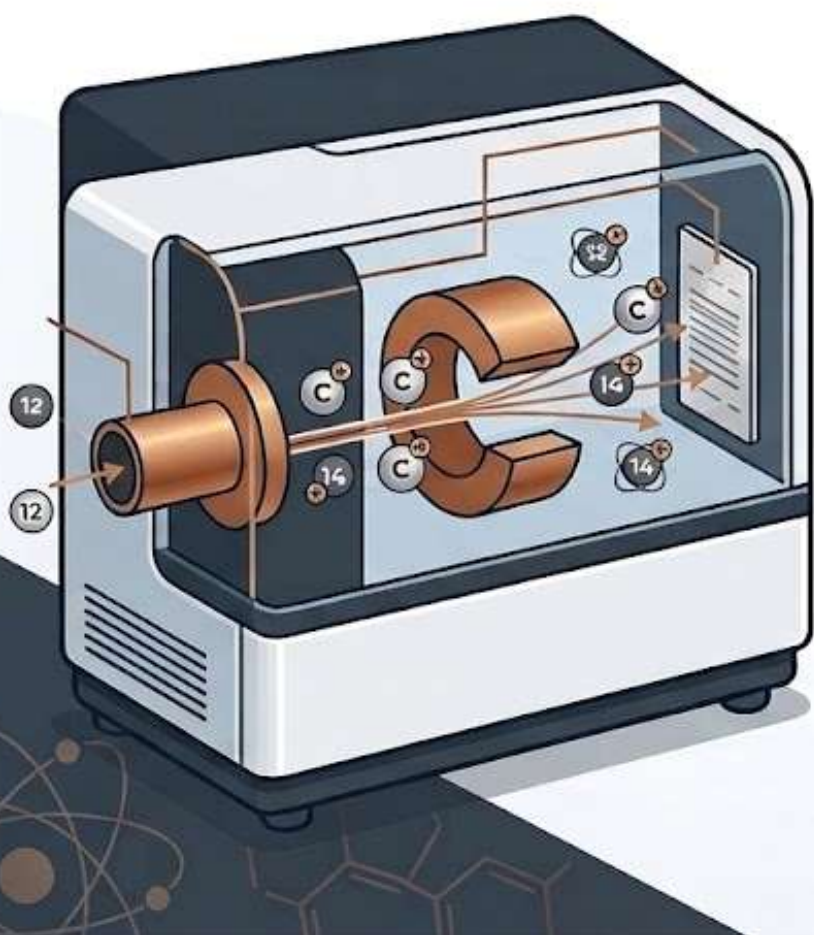
10 Mass Spectrometer

Measuring precise atomic masses and relative abundances.

The mass of an atom **cannot be measured directly.**

A mass spectrometer measures the precise masses and the relative quantity of atoms and molecules.

USES OF MASS SPECTROMETER



1 ACCURATE ATOMIC MASS DETERMINATION

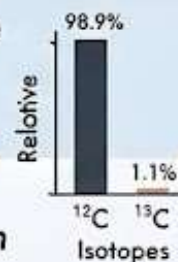
- Finds precise mass values for individual isotopes.

e.g., distinguishing ^{12}C (12.0000 amu) from ^{13}C (13.0033 amu) with tiny atomic models.

2 ISOTOPIC COMPOSITION ANALYSIS

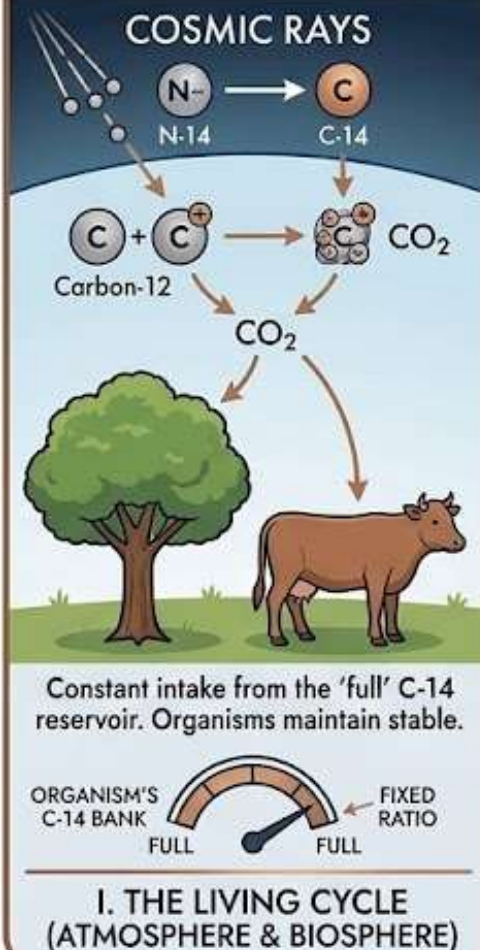
- Determines relative abundances of natural isotopes.

Used for atomic mass calculations and carbon dating.

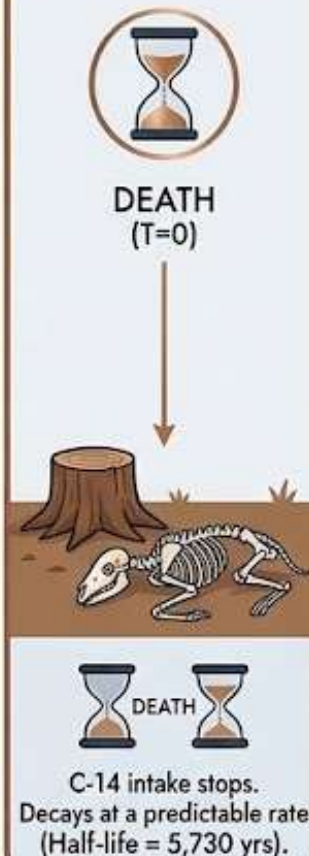


EXPLORING CARBON-14 DATING: THE RADIOACTIVE CLOCK

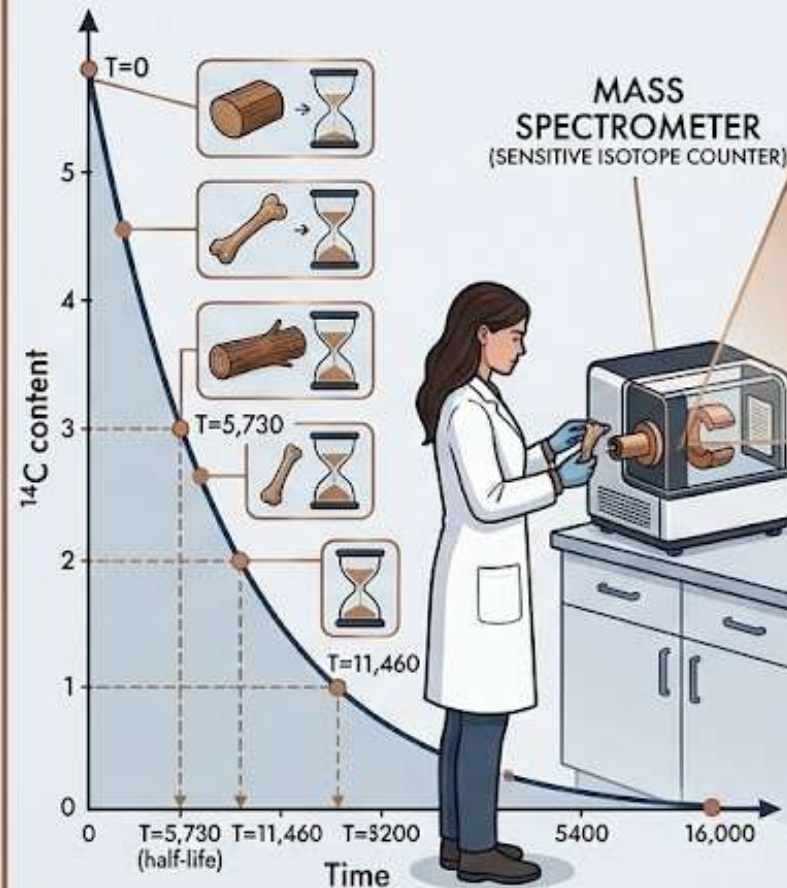
I. THE LIVING CYCLE (ATMOSPHERE & BIOSPHERE)



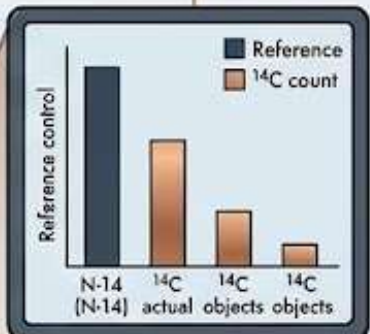
II. THE EVENT OF DEATH (CLOCK STARTS)



III. MEASURING THE RESIDUAL (TIME & ANALYSIS)



IV. CALCULATING AGE



Precise count of remaining ^{14}C relative to stable isotopes

IV. CALCULATING AGE



^{14}C Count = X%
Age = Y years



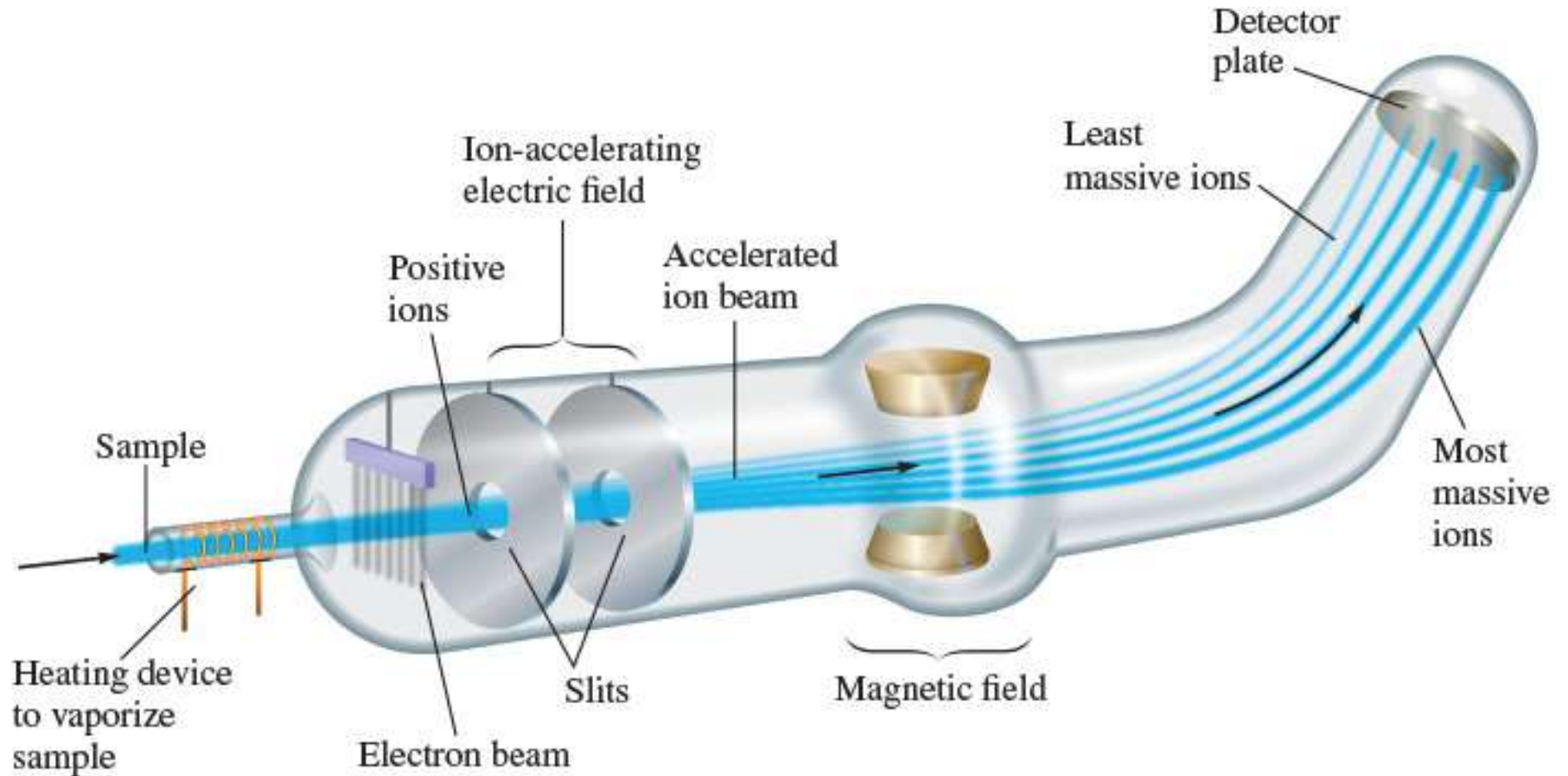
Egyptian papyrus
Age = Y years



Prehistoric bone
 ^{14}C Count = X%
Age = Y years

10. MASS SPECTROMETER

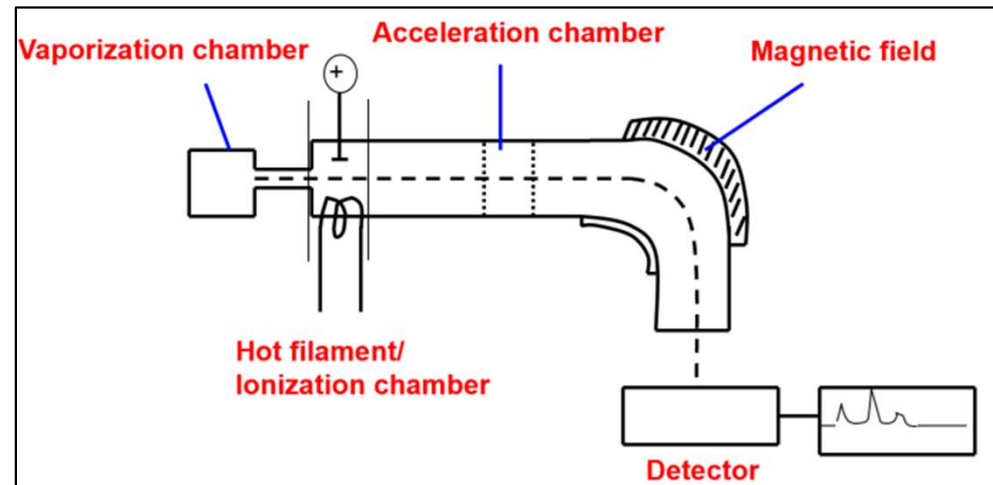
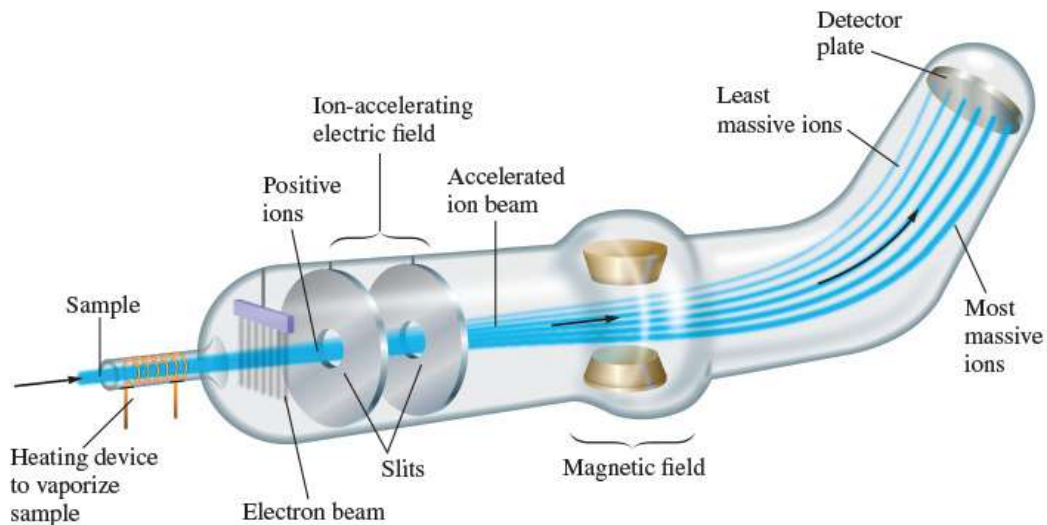
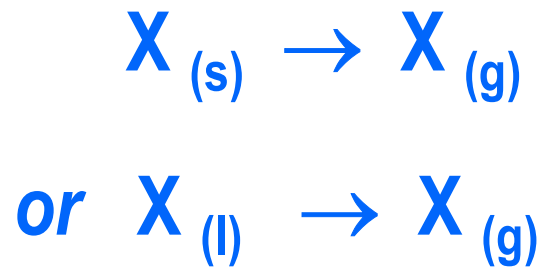
- ➡ The mass of an atom cannot be measured directly.
- ➡ A mass spectrometer is an instrument used to measure the precise masses and relative quantity of atoms and molecules.



Schematic diagram: A mass spectrometer shows a large glass tube curving upward and closed at one end.

MASS SPECTROMETER – 5 STAGES

- Once the sample of an element has been placed in the mass spectrometer, it undergoes five stages.
- Vaporisation** – the sample has to be in gaseous form. If the sample is a solid or liquid, a heater is used to vaporise some of the sample.

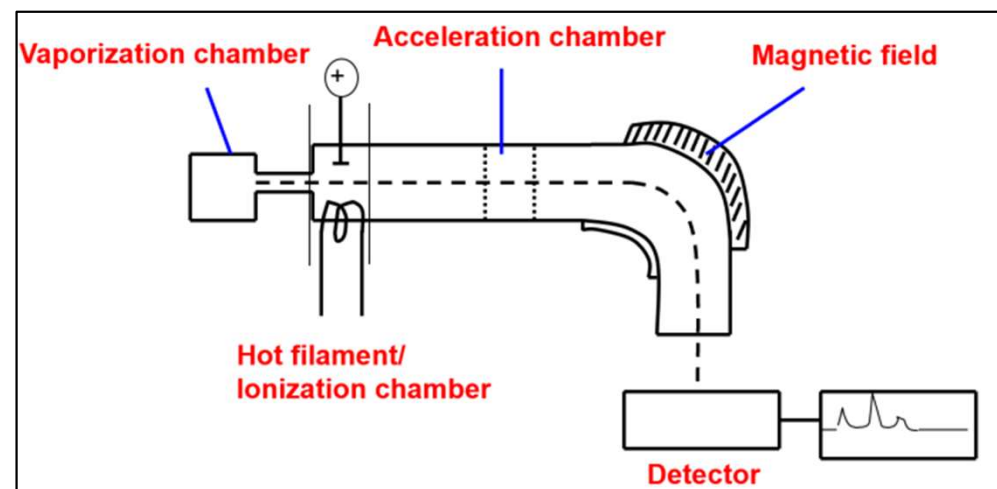
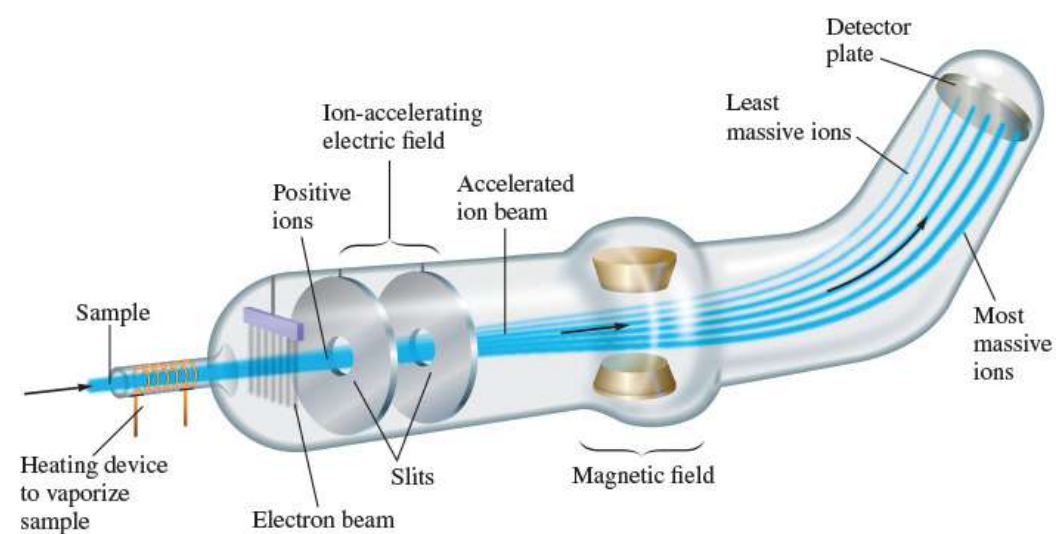


MASS SPECTROMETER – 5 STAGES

- **Ionization** – sample is bombarded by a stream of high-energy electrons from an electron gun, which ‘knock’ an electron from an atom. This produces a positive ion:

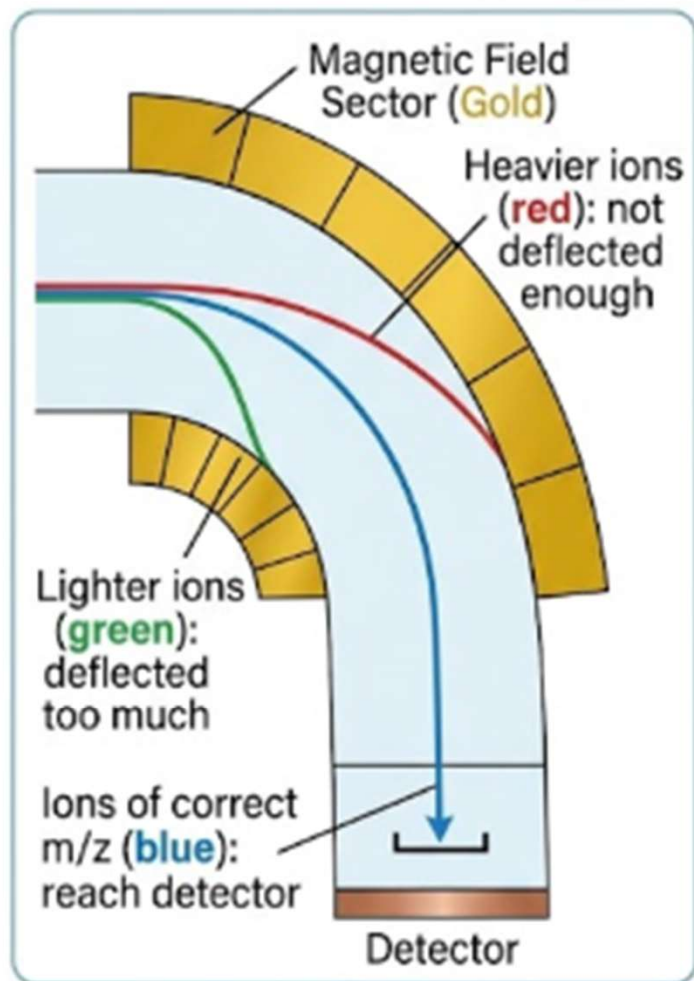


- **Acceleration** – an electric field is used to accelerate the positive ions towards the magnetic field. The accelerated ions are focused and passed through a slit: this produces a narrow beam of ions.



MASS SPECTROMETER – 5 STAGES

- **Deflection** – The accelerated ions are deflected by the magnetic field.



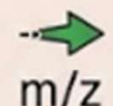
The Amount of Deflection is Greater When:



mass of positive ion is **less**



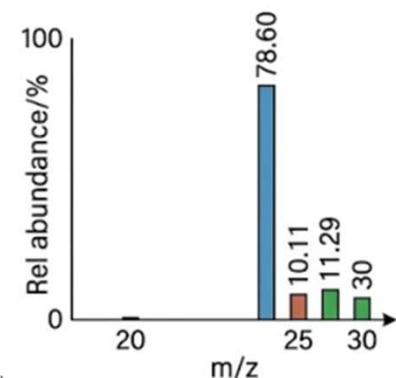
charge on positive ion is **greater**



lighter ions have **higher velocities** & lower m/z



magnetic field strength is **greater**



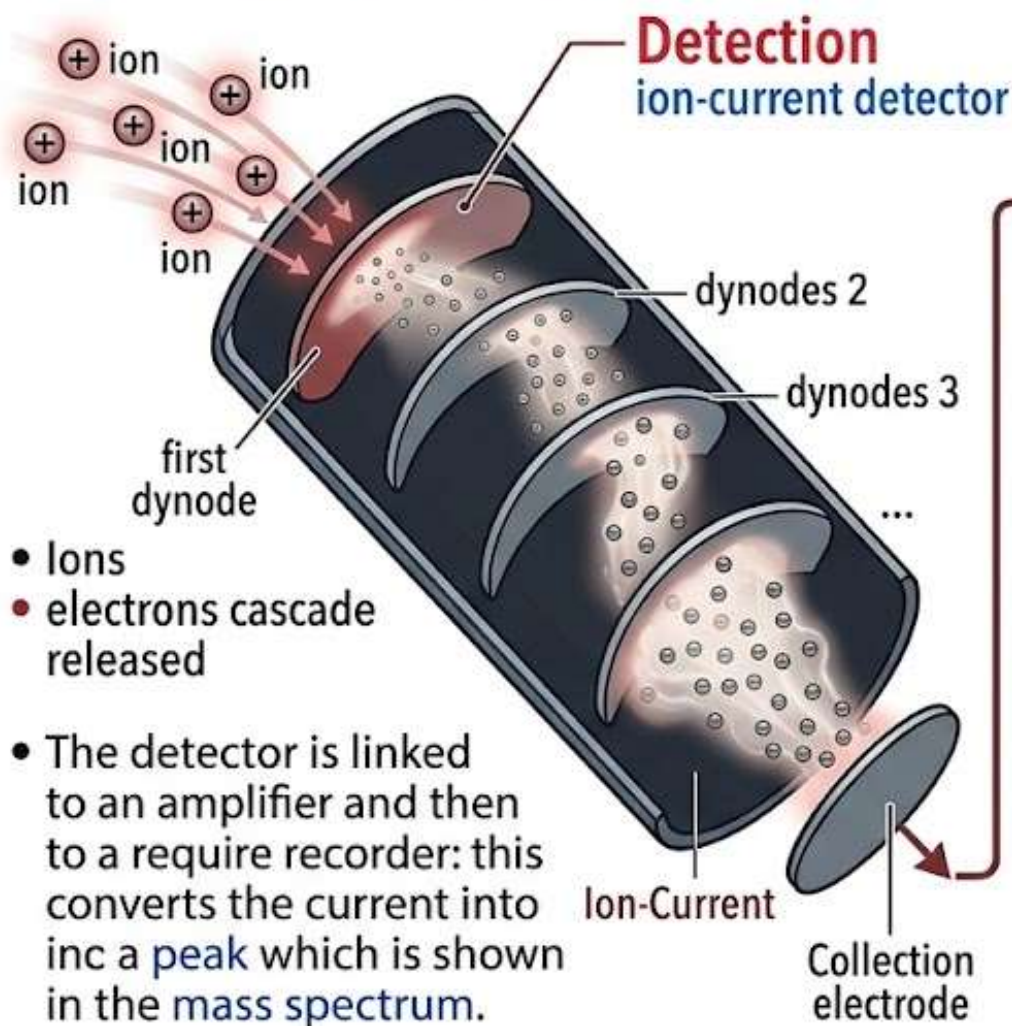
m/z

For a given magnetic field... only ions with a particular **relative mass (m) to charge (z) ratio** – the **m/z value** – are deflected sufficiently to reach the detector.

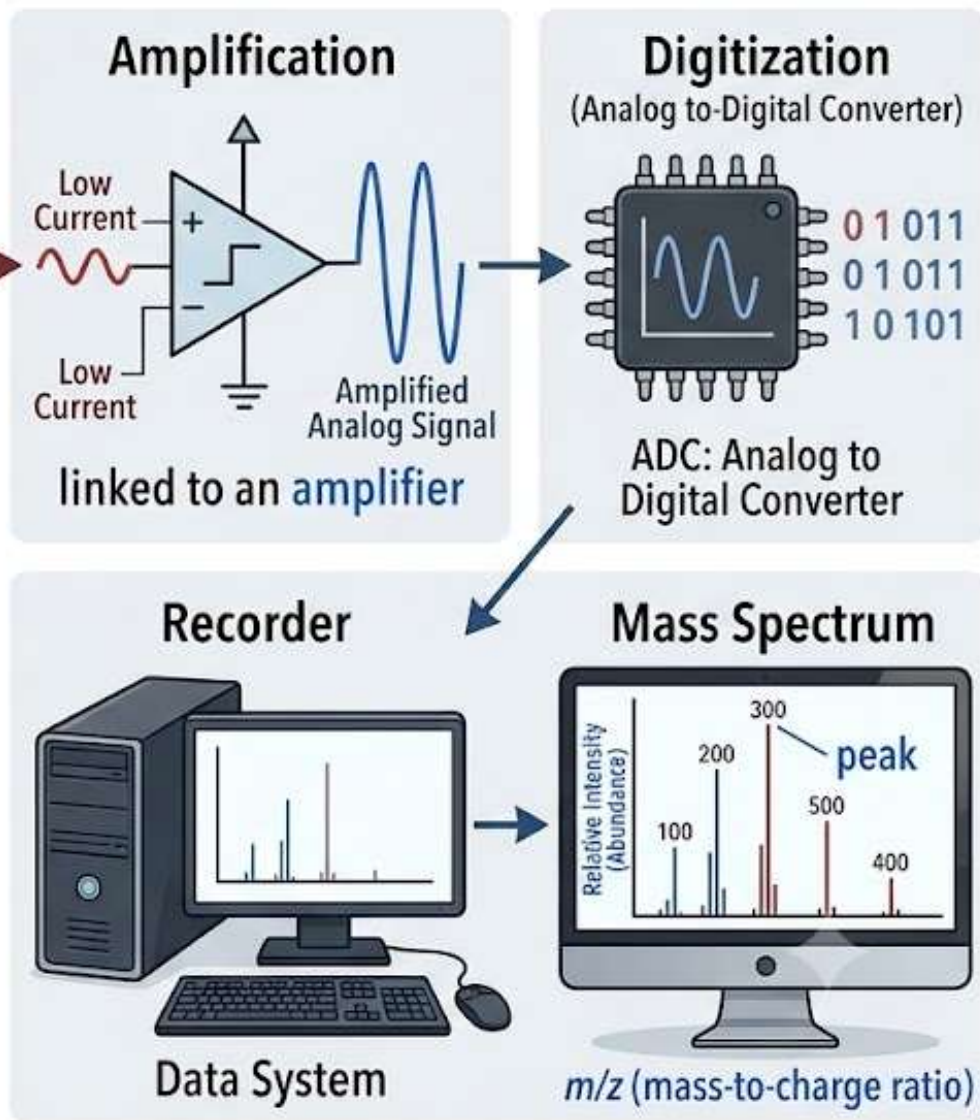
If all ions travel at the **same velocity** and carry the **same charge**... the amount of deflection depends upon the mass of the ion.

MASS SPECTROMETER: DETECTION & DATA CONVERSION

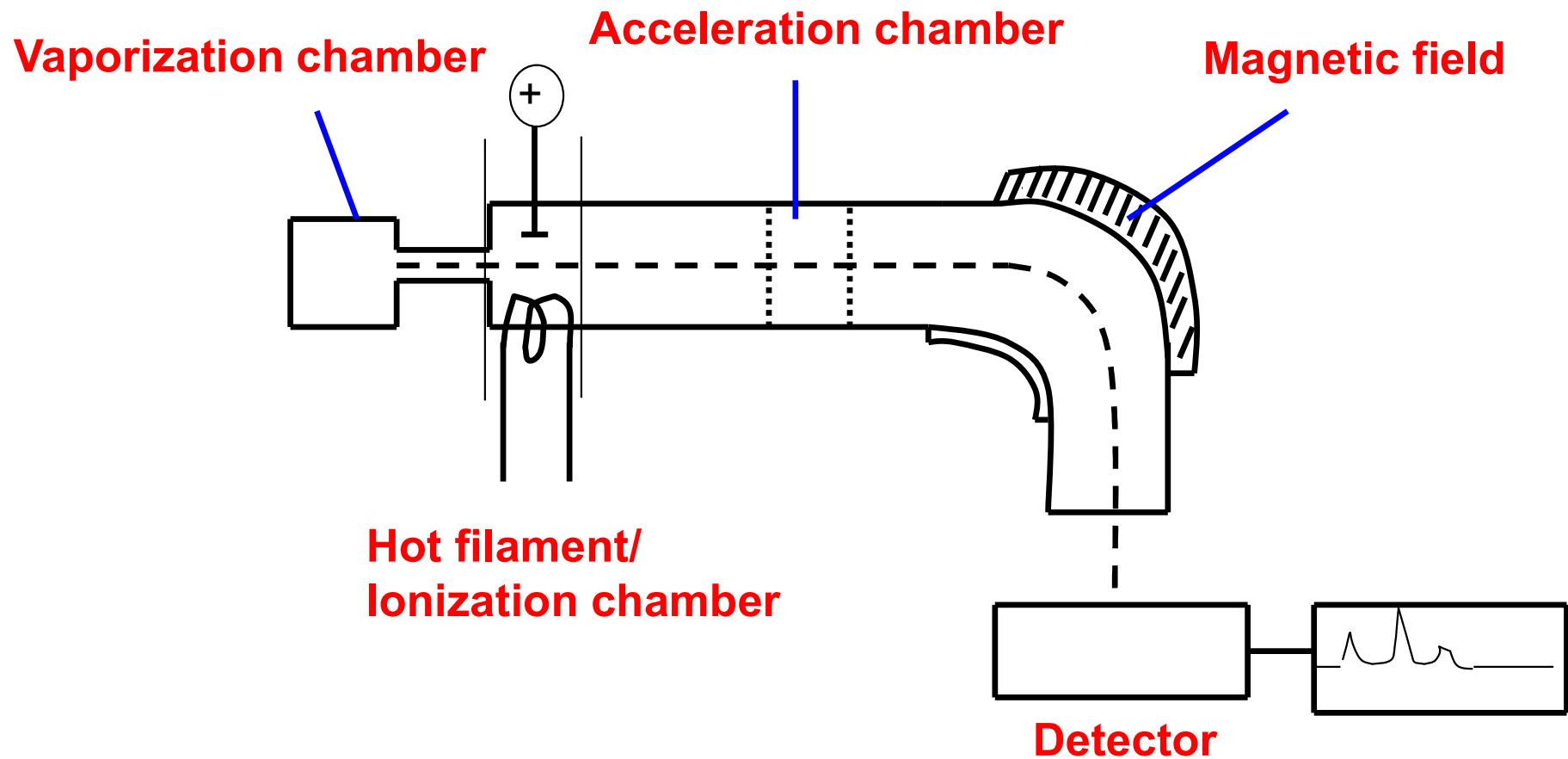
STAGE 1: ION DETECTION AT THE ION-CURRENT DETECTOR



STAGE 2: DATA PROCESSING



Schematic Diagram Of Mass Spectrometer

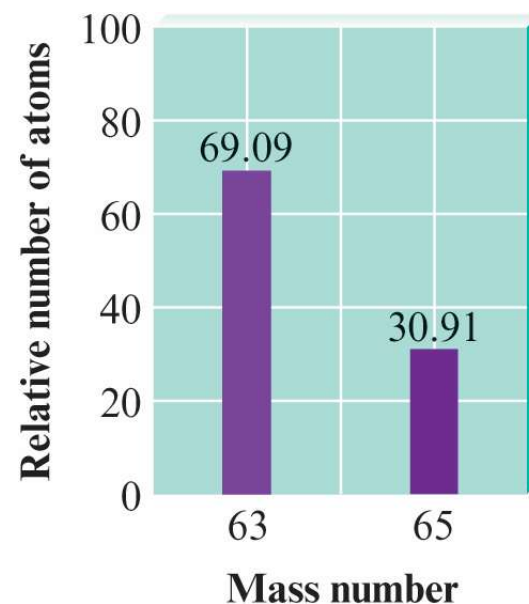


In Summary, FYI

- The relative mass of atoms is determined by mass spectrometer.
- **Vaporization chamber** the sample is vaporized into gaseous atoms.
- **Ionization chamber** the gaseous atoms are ionized by a hot wire to become gaseous positive ions.
- **Acceleration chamber** the positive ions are accelerated by an electric field towards plate .
- **Magnetic field** the positive ions are then deflected. ions with small masses are deflected the most.
- The data will be printed as a mass spectrum of the element.
- The RAM of the ion can be calculated.

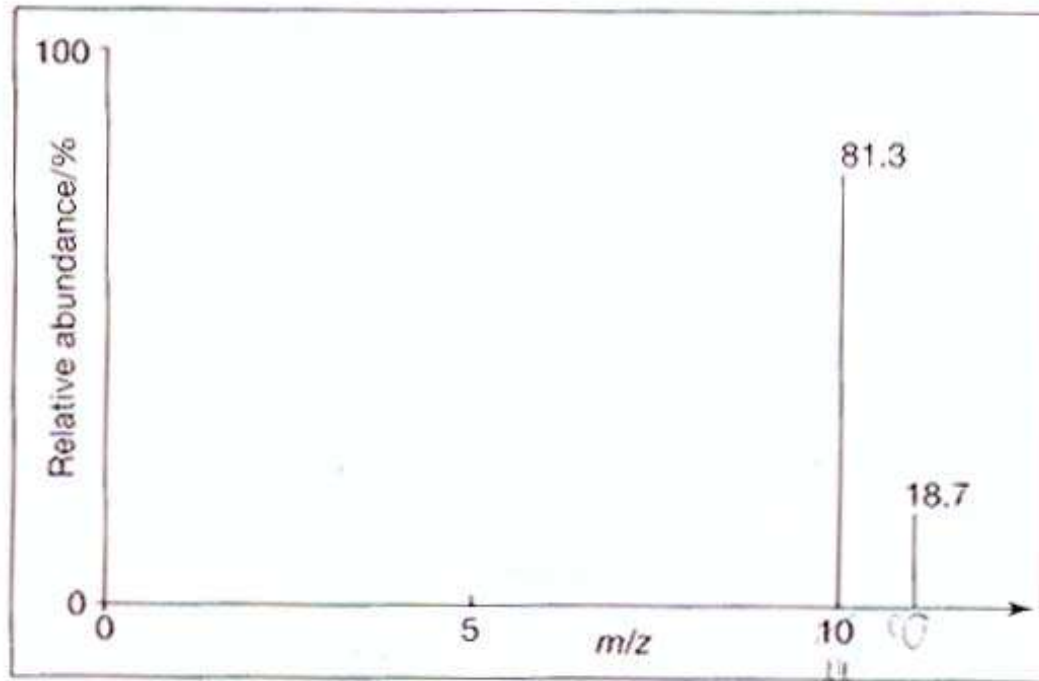
EXAMPLE : AVERAGE MASS OF AN ELEMENT

- When a sample of natural copper is vaporized and injected into a mass spectrometer, the results shown in the graph are obtained
 - Use these data to compute the average mass of natural copper
 - Mass values for ^{63}Cu and ^{65}Cu are 62.93 u and 64.93 u, respectively



Exp 1: Mass Spectrometer

☀ Isotopes of boron



m/z value	11	10
Relative abundance %	18.7	81.3

Exp 2: Mass Spectrometer

Bromine has two naturally occurring isotopes with the following percentage natural abundances:

Isotope	Natural abundance /%
^{79}Br	50.69
^{81}Br	49.31

Calculate the relative atomic mass of bromine correct to two decimal places.

Exp 3: Mass Spectrometer

- Natural carbon is a mixture of ^{12}C , ^{13}C , and ^{14}C
 - Atomic mass of carbon is an average value of these three isotopes
- Composition of natural carbon:
 - ^{12}C atoms (mass = 12 u) - 98.89%
 - ^{13}C atoms (mass = 13.003355 u) - 1.11%

Exp 4: Mass Spectrometer

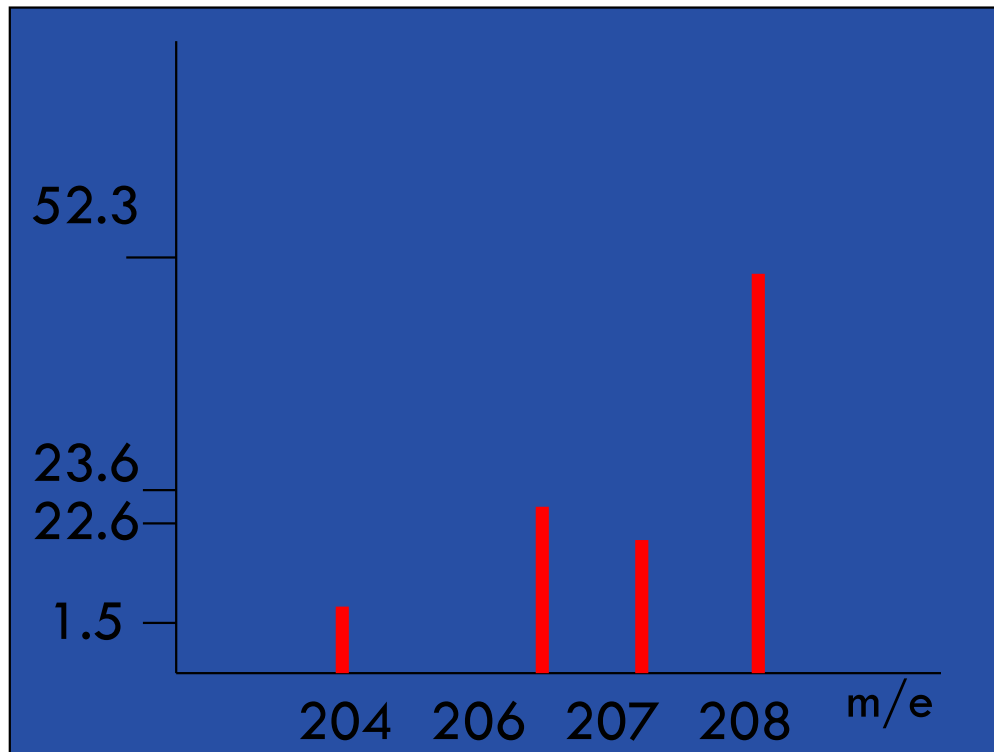
MASS SPECTROMETER – QUESTIONS

- A MASS SPEC CHART FOR A SAMPLE OF NEON SHOWS THAT IT CONTAINS:
 - 90.9% ^{20}Ne
 - 0.17% ^{21}Ne
 - 8.93% ^{22}Ne

CALCULATE THE RELATIVE ATOMIC MASS OF NEON

YOU MUST SHOW ALL YOUR WORKING!

Exp 5: Mass Spectrometer



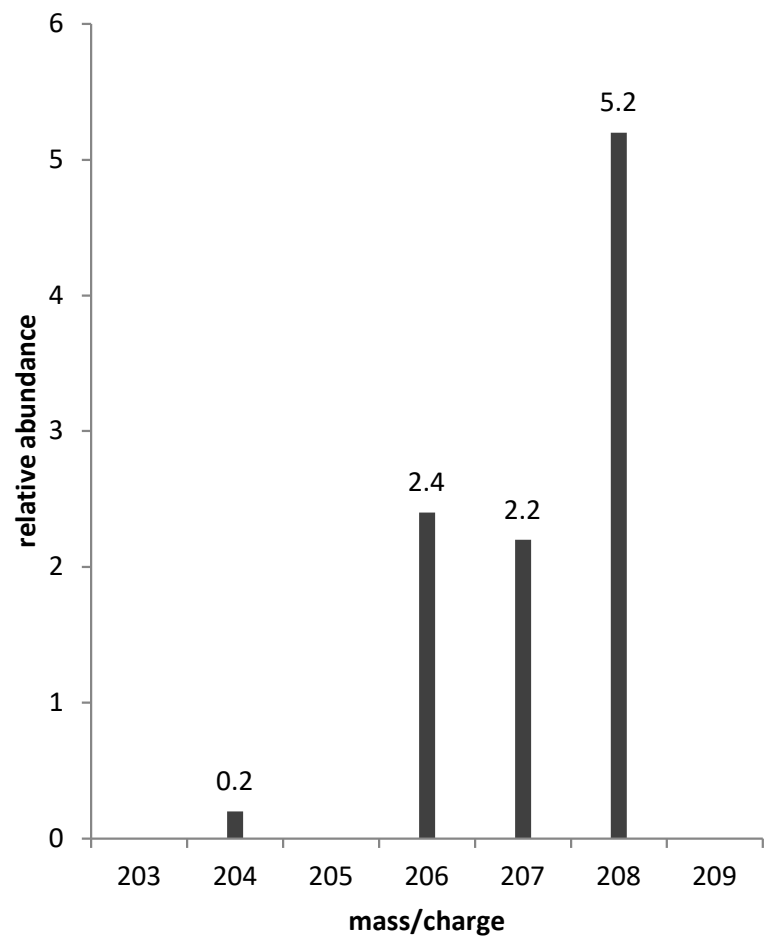
CALCULATE THE RELATIVE
ATOMIC MASS OF LEAD

YOU MUST SHOW ALL YOUR
WORKING!

Exp 6: Mass Spectrometer

EXAMPLE: FIND THE RELATIVE ATOMIC MASS (A_R) OF NATURALLY OCCURRING LEAD FROM THE DATA BELOW. RECORD YOUR ANSWER TO THE NEAREST TENTH.

Mass Spectrum of Pb

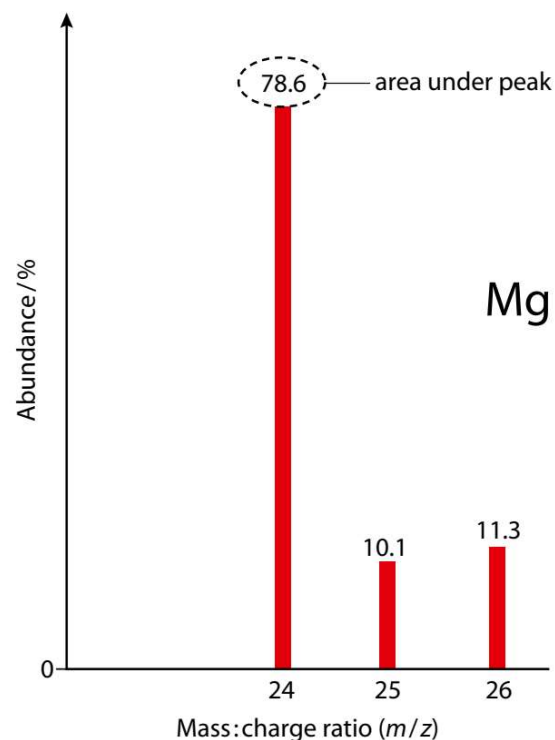


Isotopic mass	Relative abundance	% relative abundance
204	0.2	2
206	2.4	24
207	2.2	22
208	5.2	52

Figure: The Mass Spectrum of Naturally Occurring Lead

Exp 7: Mass Spectrometer

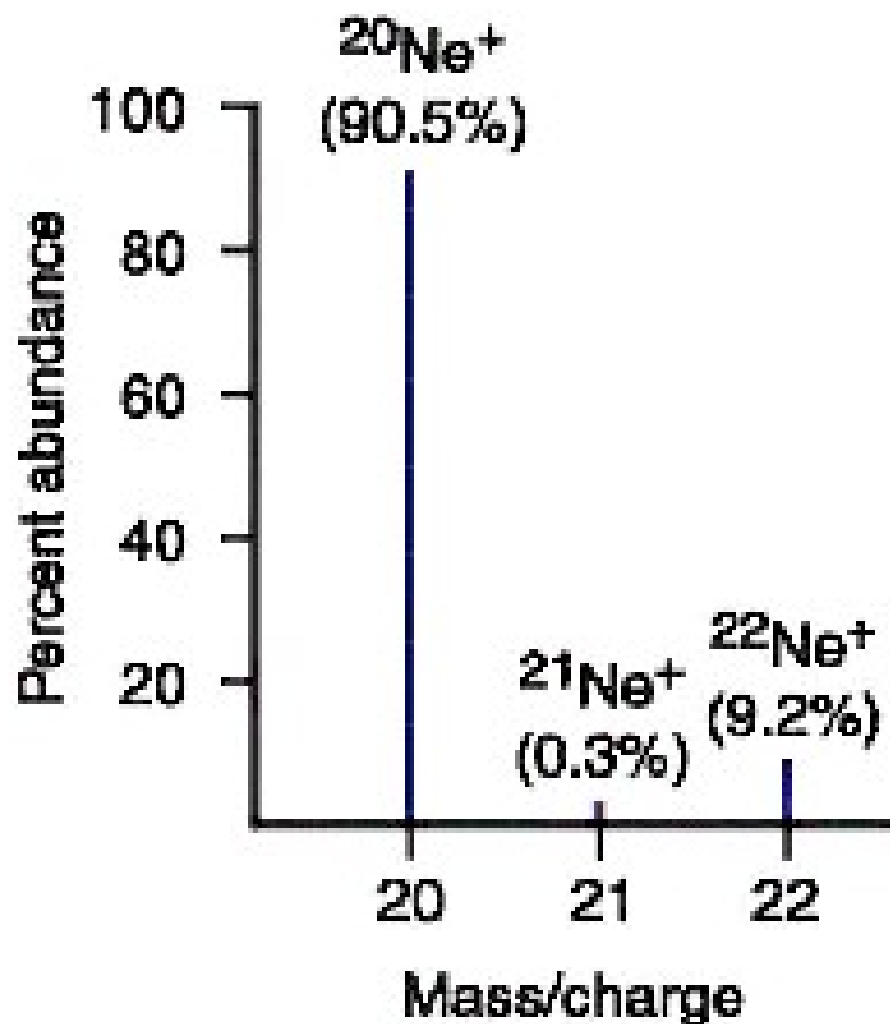
Mass Spectrum of Magnesium



- The mass spectrum of Mg shows that Mg consists of 3 isotopes: ^{24}Mg , ^{25}Mg and ^{26}Mg .
- The height of each line is proportional to the abundance of each isotope.
- ^{24}Mg is the most abundant of the 3 isotopes.

Question 1:

Calculate the relative atomic mass of neon from the mass spectrum.



Question 2:

Copper occurs naturally as mixture of 69.09% of ^{63}Cu and 30.91% of ^{65}Cu . The isotopic masses of ^{63}Cu and ^{65}Cu are 62.93 amu and 64.93 amu respectively. Calculate the relative atomic mass of copper.

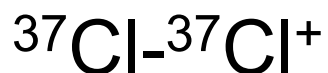
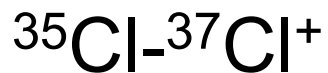
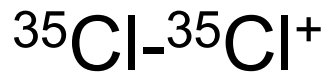
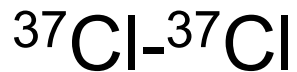
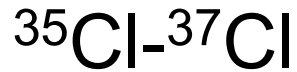
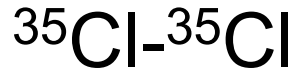
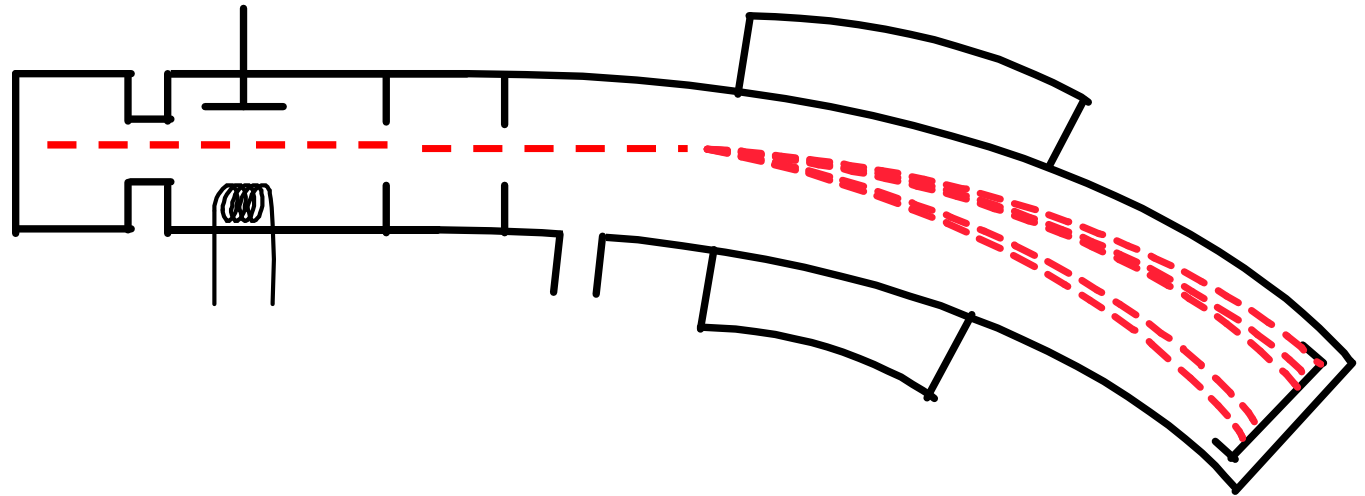
Question 3:

Naturally occurring iridium, Ir is composed of two isotopes, ^{191}Ir and ^{193}Ir in the ratio of 5:8. The relative isotopic mass of ^{191}Ir and ^{193}Ir are 191.021 amu and 193.025 amu respectively. Calculate the relative atomic mass of Iridium.

Mass Spectrum of Molecular Elements

A sample of chlorine which contains 2 isotopes with nucleon number 35 and 37 is analyzed in a mass spectrometer. how many peaks would be expected in the mass spectrum of chlorine?

MASS SPECTROMETER

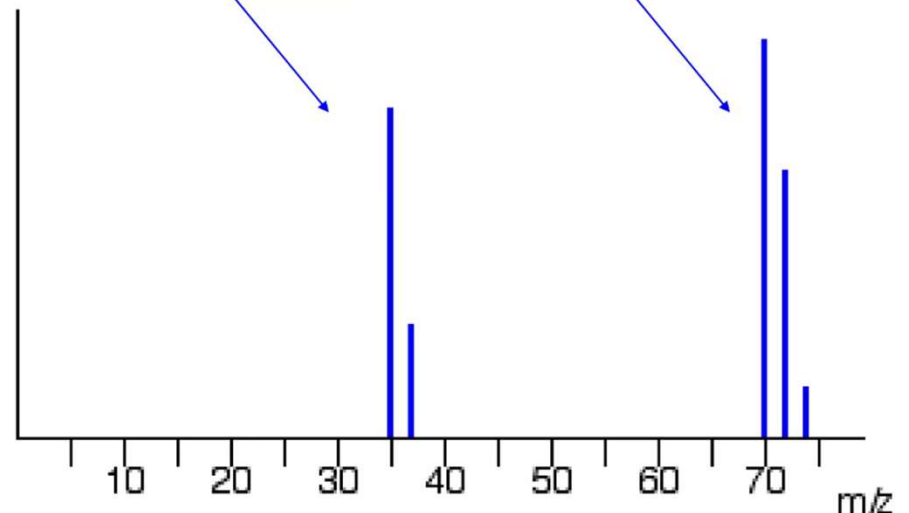


A sample of chlorine which contains 2 isotopes with nucleon number 35 and 37 is analyzed in a mass spectrometer. how many peaks would be expected in the mass spectrum of chlorine?

Isotopes Cl-35 & Cl-37

Molecules Cl_2 -70 Cl_2 -72 Cl_2 -74

relative abundance



Ratio $^{35}\text{Cl}^+ : ^{37}\text{Cl}^+ = 3:1$

Ratio $^{35}\text{Cl}-^{35}\text{Cl}^+ : ^{35}\text{Cl}-^{37}\text{Cl}^+ : ^{37}\text{Cl}-^{37}\text{Cl}^+ = 9:6:1$



Ratio $^{35}\text{Cl}^+ : ^{37}\text{Cl}^+ = 3:1$

Ratio $^{35}\text{Cl}-^{35}\text{Cl}^+ : ^{35}\text{Cl}-^{37}\text{Cl}^+ : ^{37}\text{Cl}-^{37}\text{Cl}^+ = 9:6:1$

